# FloodLAMP

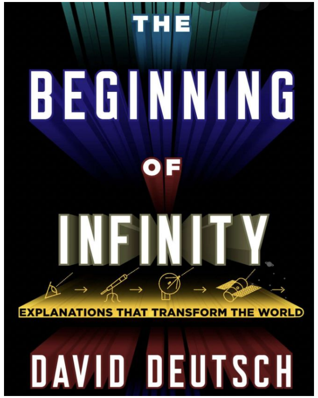
## Biotechnologies, PBC

**Extending the Reach of LAMP:**
the FloodLAMP platform for globally accessible disease screening

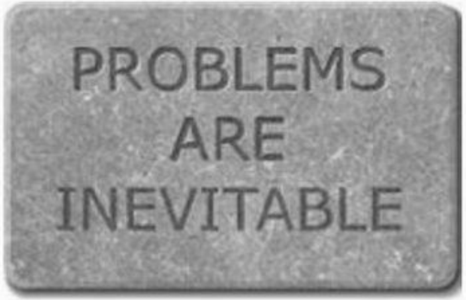
New England Biolabs Seminar - March 3, 2022

**Randy True,** Founder and CEO | randy@floodlamp.bio

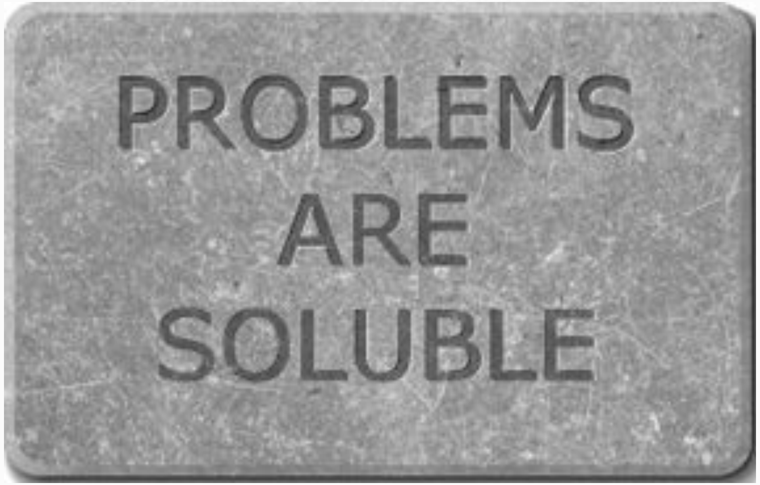
# David Deutsch's Deep Optimism:




a much more general truth, namely that, for people, *problems are inevitable*. So let us carve *that* in stone:




It is inevitable that we face problems, but no particular problem is inevitable. We survive, and thrive, by solving each problem as it comes up. And, since the human ability to transform nature is limited only by the laws of physics, none of the endless stream of problems will ever constitute an impassable barrier. So a complementary and equally important truth about people and the physical world is that *problems are soluble*. By 'soluble' I mean that the right knowledge would solve them. It is not, of course, that we can possess knowledge just by wishing for it; but it is in principle accessible to us. So let us carve that in stone too:



# We *still* need better testing!

The Coronavirus Pandemic ›

LIVE Covid-19 Updates

## All Anyone Wants for Christmas Is a Covid Test


## Covid has created a capitalist nightmare

Big Tech and Big Pharma are more powerful than ever

The New York Times

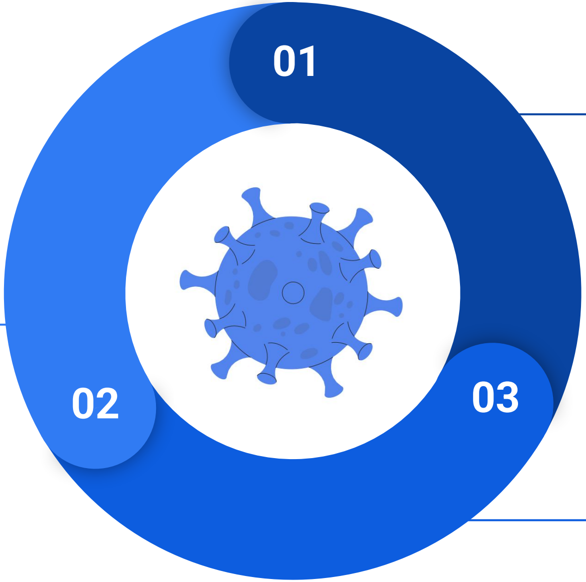
## *Biden Promised 500 Million Free Covid Tests. Then He Had to Find Them.*

Over the past two years, as the pandemic claimed the lives of millions of people and upturned the lives of everyone else, a silent revolution was taking place. Western capitalism suddenly found itself usurped; replaced by an even more oligarchic and authoritarian capitalist mode of power — what we might call neo-feudalism. Even as restrictions are lifted in the UK and other countries, the consequences of these changes will be felt for a long time to come.

On the one hand, hundreds of thousands of small and medium businesses were forced to shut down, triggering one of the worst jobs crises since the Great Depression. In 2020 alone, workers around the world cumulatively lost $3.7 trillion in earnings — an 8.3% decline.

## Demand for Covid-19 testing is falling, but experts caution it's as important as ever

*"FloodLAMP's mission is to improve global health and resiliency through universal access to rapid molecular testing."*




**New testing technology**

**Turnkey screening programs**

**Disruptive open source strategy**

# 1) Real World Programs

# Operating Programs in the Real World

## EMS Leadership Conference Screening

- 1,000 attendees

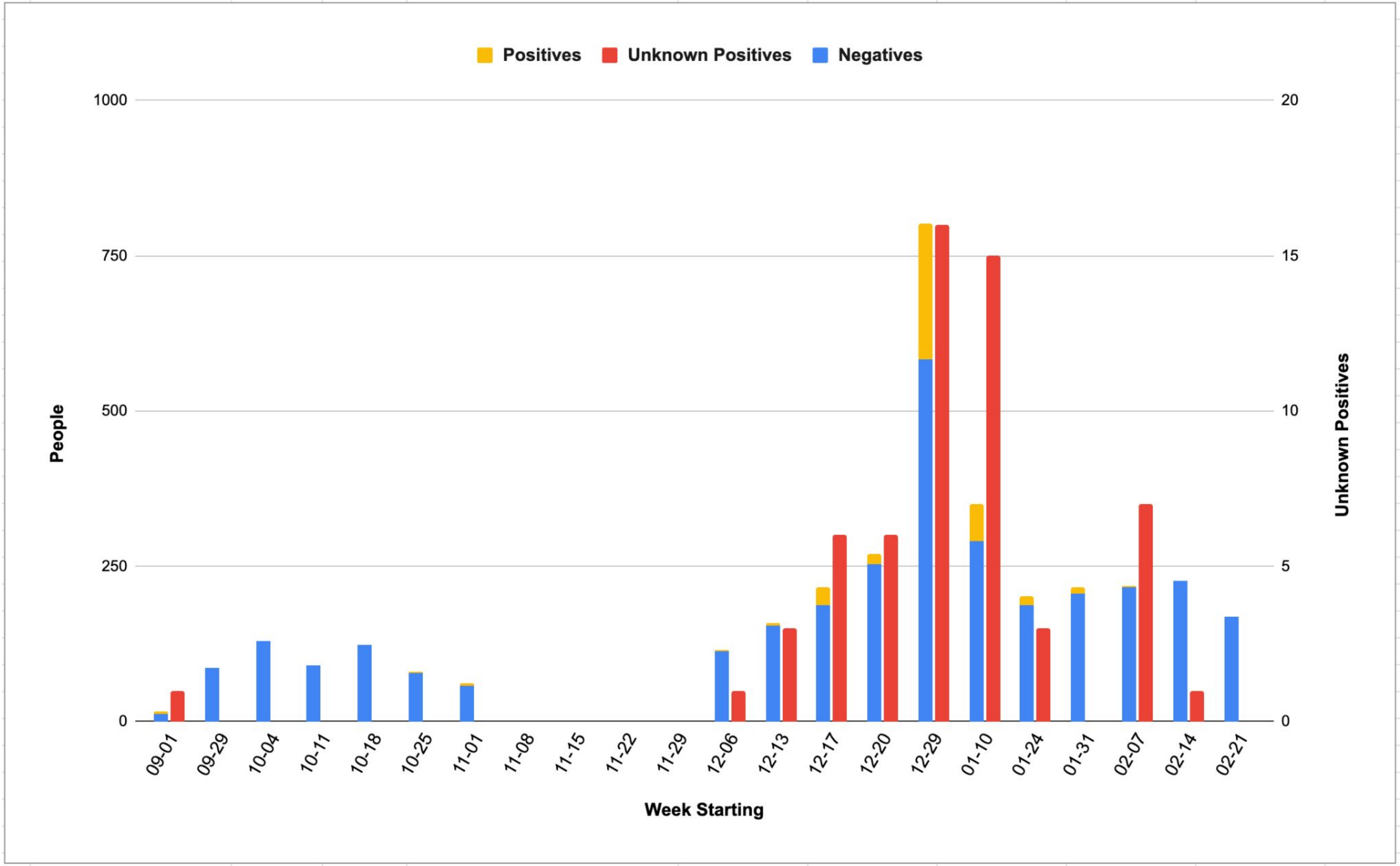

## Municipal & EMS Deployments

- Coral Springs, FL — approx. 132,000
- Davie, FL — approx. 104,000
- Bend, OR — approx. 94,000

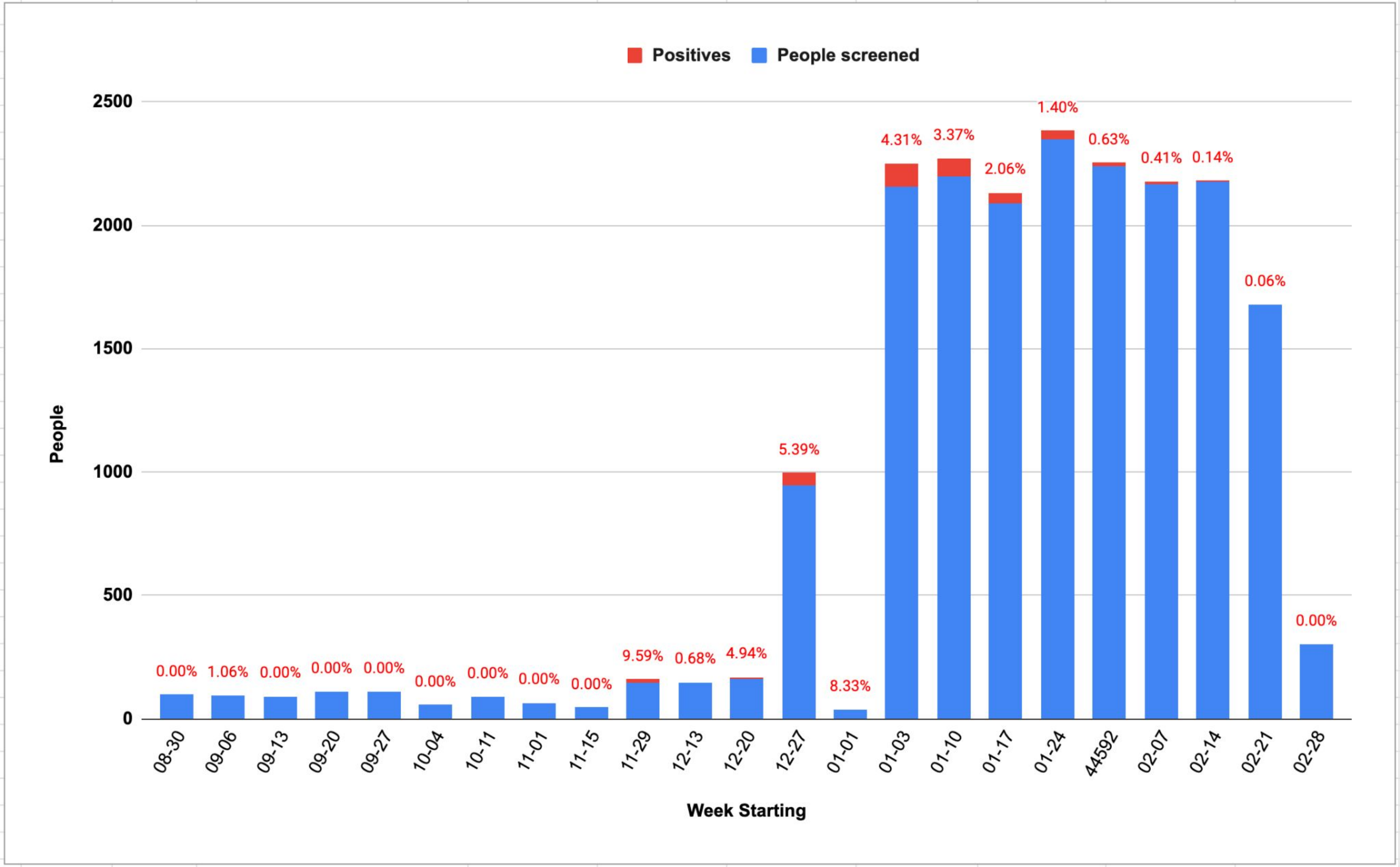
## Preschool Screening with Family Pooling

- Approx. 40 families
- 65-105 people per bi-weekly screening

# Town of Davie, FL



# Coral Springs, FL



# Coral Springs, FL



This novice (but dedicated) tester is able to screen 1000 critical city employees.

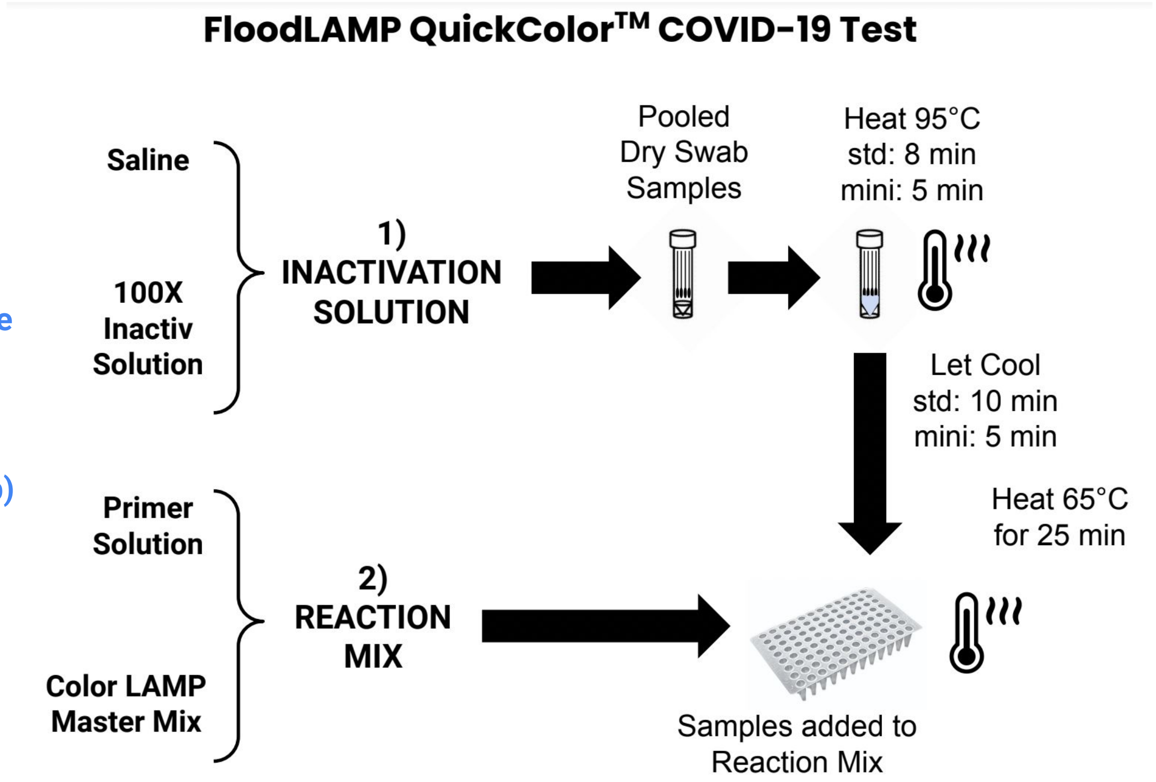
# Davie, FL

# Test Diagrams

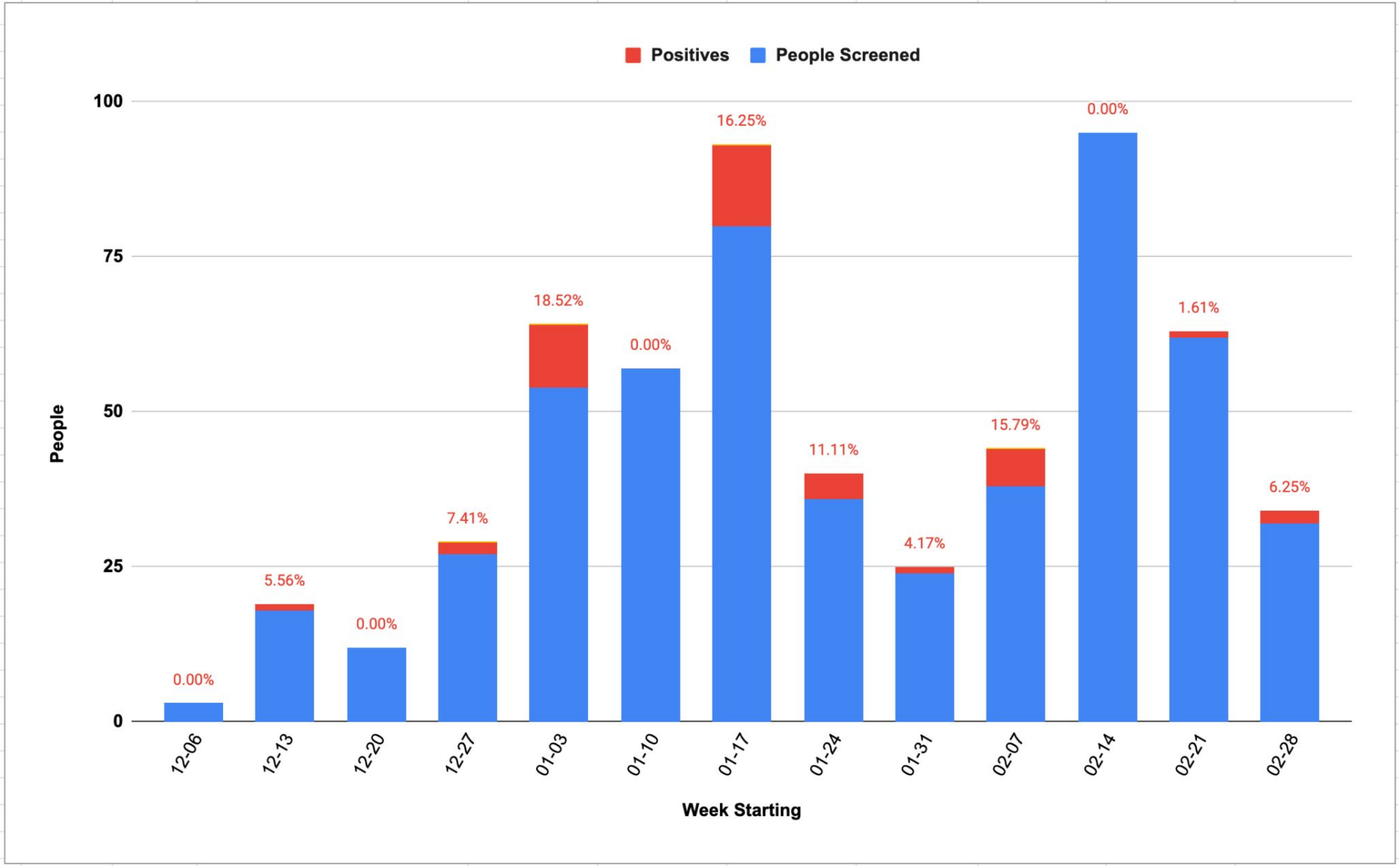
TCEP/EDTA
(standard Rabe
Cepko)

AS1e (ORF1ab)
N2
E1

NEB M1804

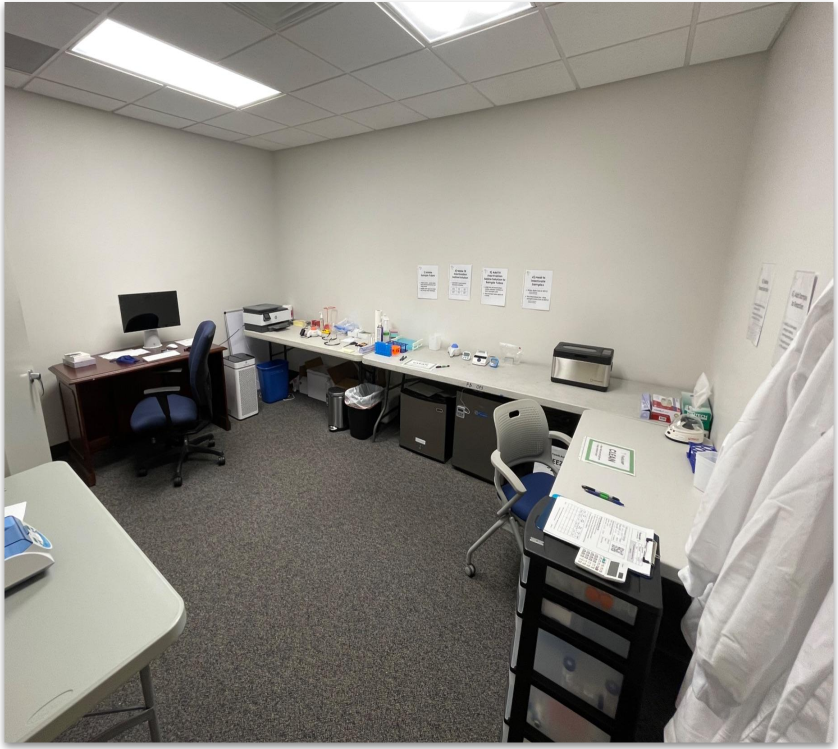


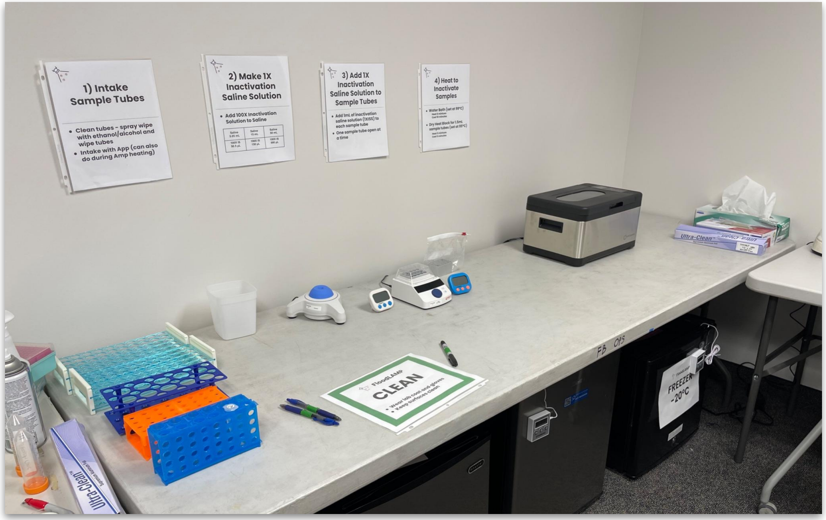
# Bend, OR


Positives
People Screened
100
75
50
25
0
People
0.00%
5.56%
0.00%
7.41%
18.52%
0.00%
16.25%
11.11%
4.17%
15.79%
0.00%
1.61%
6.25%
12-06
12-13
12-20
12-27
01-03
01-10
01-17
01-24
01-31
02-07
02-14
02-21
02-28
Week Starting

# Nova, Southeastern University Lab

Lab setup (Jan 2022)





# FloodLAMP accepted into Levan Innovation Center


# Early Childcare COVID-19 Family Screening Pilot: Carillon Preschool

Protecting young children during the Omicron surge

Prepared by:

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**Theresa Ling** – User Experience and Design, FloodLAMP Biotechnologies, PBC
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**FloodLAMP Biotechnologies**, a Public Benefit Corporation    floodlamp.bio    1

*"I really appreciate this process. D tested positive on antigen test at home today and is now showing symptoms, so I truly appreciate the surveillance program picking this up! Thank you again!!!"*
- Parent of a student who was referred to follow up testing while negative on antigen and pre-symptomatic

*"Thank you so much for helping to get our kids tested and doing what you can to keep them safely in school! We are all so lucky to have you as part of our community!"*
- Parent of a student and MD, wrote to school administrators to move to mandatory testing with FloodLAMP

*"I thought [FloodLAMP] was perfect. Easy and convenient."*
- Parent of a student and restaurant owner who later wanted to extend FloodLAMP testing to her restaurant employees

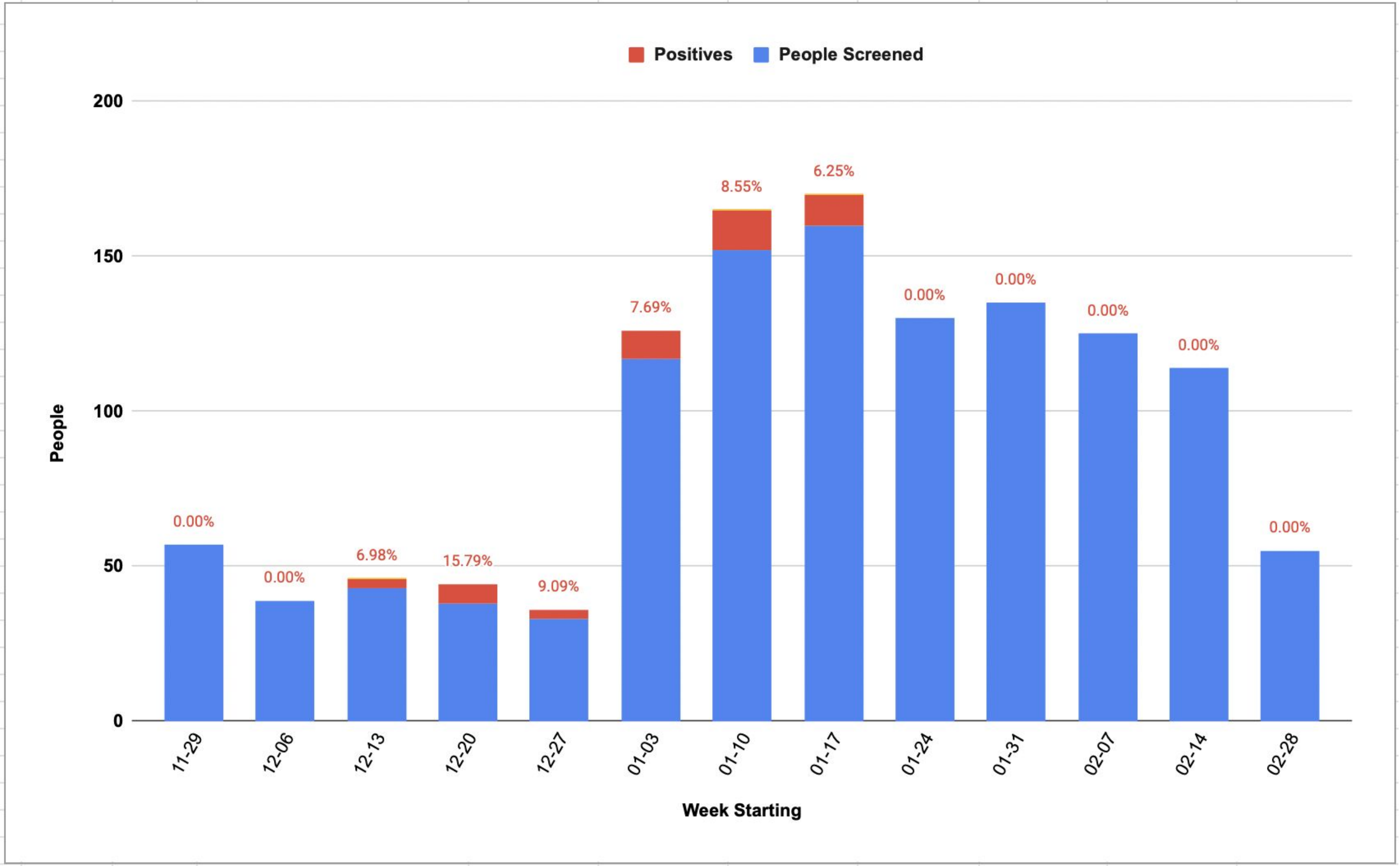
*"You are a godsend to our community. I cannot begin to thank you for all of your support and love in keeping our community safe in this difficult time."*
- Preschool teacher and administrator

## Link for Preschool Pilot Description

https://docs.google.com/document/d/1TOnklq-65XUX-v-li018rteK9jeL8NgqbNrtbndgD_o

FLOODLAMP ARCHIVE FILE PATH: various/fl-whitepapers/FloodLAMP Whitepaper - California Preschool Family Pooled Screening Pilot (June 2022).md

# Preschool, CA

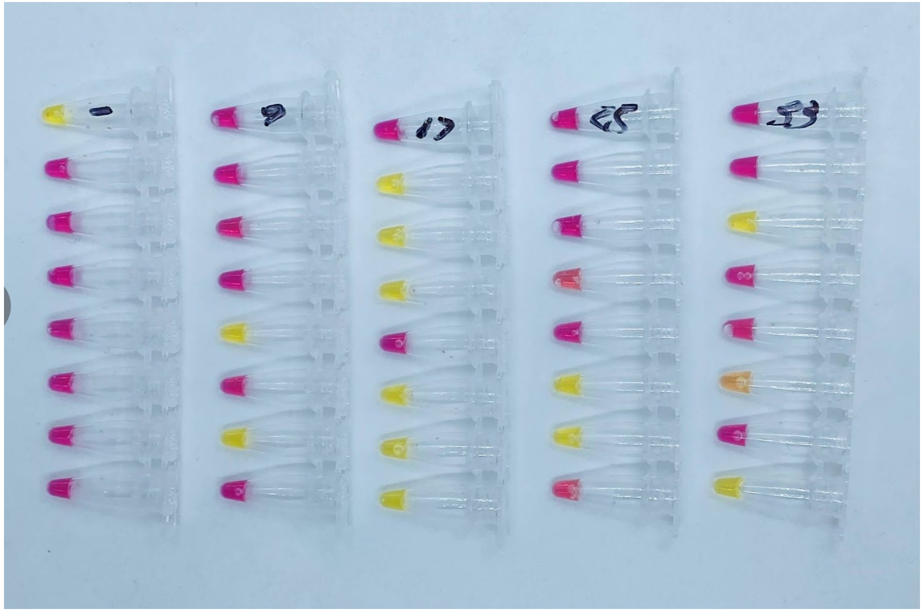


# Pre-School Screening - Case Report

- Several infections detected by FloodLAMP 1-2 days before symptoms or antigen tests
- In 3 different cases, family members of the preschool student were detected, showing the advantage of family pooling in protecting spread in a congregate setting such as schools and workplaces.

| | 1/2/22 | 1/3/22 | 1/4/22 | 1/6/22 | 1/7/22 | 1/8/22 | 1/9/22 | 1/10/22 | 1/13/22 | 1/14/22 | 1/17/22 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 4 | 5 | 6 | 7 | 8 | 11 | 12 | 15 |
| Family Pool | POS FloodLAMP | | | | | | | | | | POS FloodLAMP |
| Parent 1 | Asymptomatic | | | | | | | | | | |
| | POS FloodLAMP (Family Pool) | POS FloodLAMP (Indiv) | | | | | POS FloodLAMP (Indiv) | > POS FL Result | | | POS FloodLAMP (Family Pool) |
| | NEG Antigen | | | | | | | | | NEG Antigen | |
| | | | POS PCR Lab > | > > > > > | > POS PCR Result | | | | | | |
| Parent 2 | Asymptomatic | | Symptoms | | | | | | | | |
| | POS FloodLAMP (Family Pool) | POS FloodLAMP | | | | | POS FloodLAMP (Indiv) | > POS FL Result | | | POS FloodLAMP (Family Pool) |
| | NEG Antigen | | | | | | | | | NEG Antigen | |
| | | POS PCR Lab > | > POS PCR Result | | | | | | | | |
| Child 1 | Asymptomatic | | Symptoms | Asymptomatic | | | | | | | |
| | POS FloodLAMP (Family Pool) | NEG FloodLAMP | | | | | POS FloodLAMP (Indiv) | > POS FL Result | | | POS FloodLAMP (Family Pool) |
| | | | POS Antigen | | | | | | | NEG Antigen | |
| | | | | | POS PCR Lab > | > POS PCR Result | | | | | |
| Child 2 | Asymptomatic | | Symptoms | Asymptomatic | | | | | | | |
| | POS FloodLAMP (Family Pool) | NEG FloodLAMP | | | | | POS FloodLAMP (Indiv) | > POS FL Result | | | POS FloodLAMP (Family Pool) |
| | | | POS Antigen | | | | | | POS Antigen (slight | NEG Antigen | |
| | | | | POS PCR Lab > | > POS PCR Result | | | | | | |

# Pre-School Screening



**Mon 1-10**

**Thurs 1-13**

Mix of self collected individuals and family pools

# New Year’s Eve Case Report

- FloodLAMP caught asymptomatic infection of advisor that 3 expensive OTC/POC rapid molecular tests missed (Detect, Lucira, and Mesa Accula)
- BinaX now wasn't positive until 2 days later.
- All FloodLAMP positive samples confirmed by PCR

**FloodLAMP New Years Eve Case Report**
**caught asymptomatic infection that all antigen and OTC/POC molecular tests missed**

| | | | | | | |
|---|---|---|---|---|---|---|
| FloodLAMP EasyPCR on saliva | | | | POS - SalivaDirect PCR Ct 27.7 8 PM | POS - SalivaDirect PCR Ct 29.0 11 AM | |
| Accula Rapid Molecular PCR (Worksite) | | | | | Accula NEG - nasal swab 4 PM | |
| Lucira Rapid Molecular LAMP | | | | Lucira NEG - nasal swab 5 PM | | |
| Detect Rapid Molecular LAMP | NEG - nasal swab 10 AM | | Detect NEG - nasal swab 2 PM | | | |
| CaseStart Antigen | | | | | | |
| FlowFlex Antigen | | NEG - nasal swab 8 AM | NEG - nasal swab 2 PM | | | POS - nasal swab 10 AM |
| BinaXNow Antigen | | | | NEG - nasal swab 5 PM | NEG - nasal and throat swabs 10 AM | POS - nasal swab 10 AM |
| FloodLAMP QuickColor LAMP Nasal Swab | | NEG - nasal swab 1 PM | | NEG - nasal swab PCR Ct Und/39.4 5PM | POS - nasal swab PCR Ct 32.2 10 AM | |
| FloodLAMP QuickColor LAMP Throat Swab | | | POS - nasal swab PCR Ct 33.9 9 AM | POS - throat swab PCR Ct 30.2 5 PM | POS - throat swab PCR Ct 35.5 10 AM | |
| | 12-27-21 | 12-29-21 | 12-31-21 | 12-31-22 | 1-1-22 | 1-2-22 |

# Surveillance Testing Stats

| FloodLAMP Surveillance Testing Stats | | | Thru Mar 2, 2022 |
|---|---|---|---|
| Org | Pools | People | Positives |
| FL FF | 1,107 | 1,779 | 44 |
| FL FTFC | 61 | 195 | 0 |
| Kent Summer Camp | 190 | 696 | 0 |
| Coral Springs City | 7,400 | 21,676 | 348 |
| Davie Fire Rescue | 2,235 | 4,226 | 55 |
| Pink Shoes Production | 671 | 671 | 1 |
| Bend Fire Rescue | 617 | 617 | 215 |
| **TOTAL** | **12,281** | **29,860** | **663** |
| | | | |
| **Avg Pool Size** | **2.4** | | |

# 2) Platform - Wrap Around Components

# 2) Platform – Wraparound program components

## Applications

- COVID Response
- Emerging Threats (Pandemic Prep)
- TB, Zikka
- Influenza, STD's, Cancer

## Core Modules

**PHYSICAL**

- Reagent Test Kits
- Collection Kits
- Standard Consumables
- Standard Equipment

**DIGITAL**

- App
- Admin Portal
- Training Program
- Quality Management System

## Network

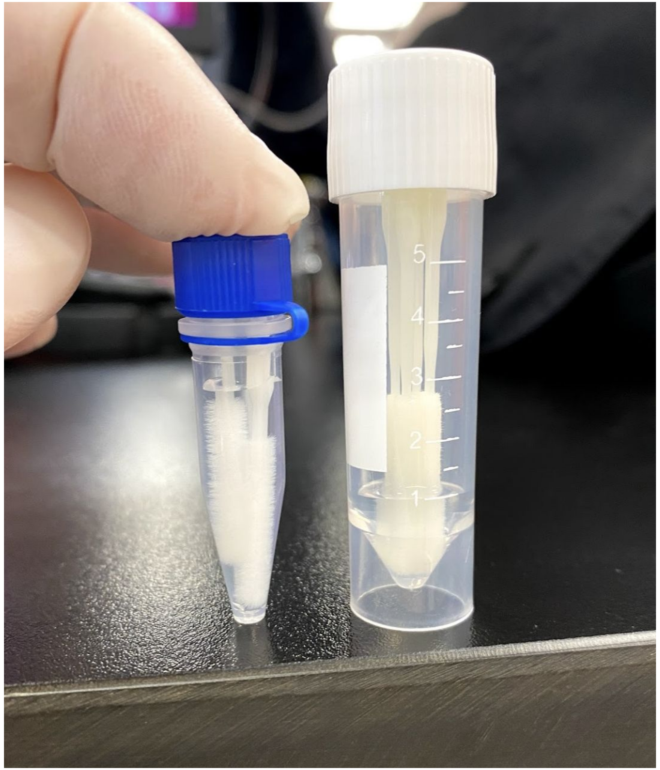
- EMS & Municipalities
- Schools and Businesses
- U.S. Federal Agencies (BARDA, ASPR, CDC, DoD, FEMA)
- International Governments

## Products

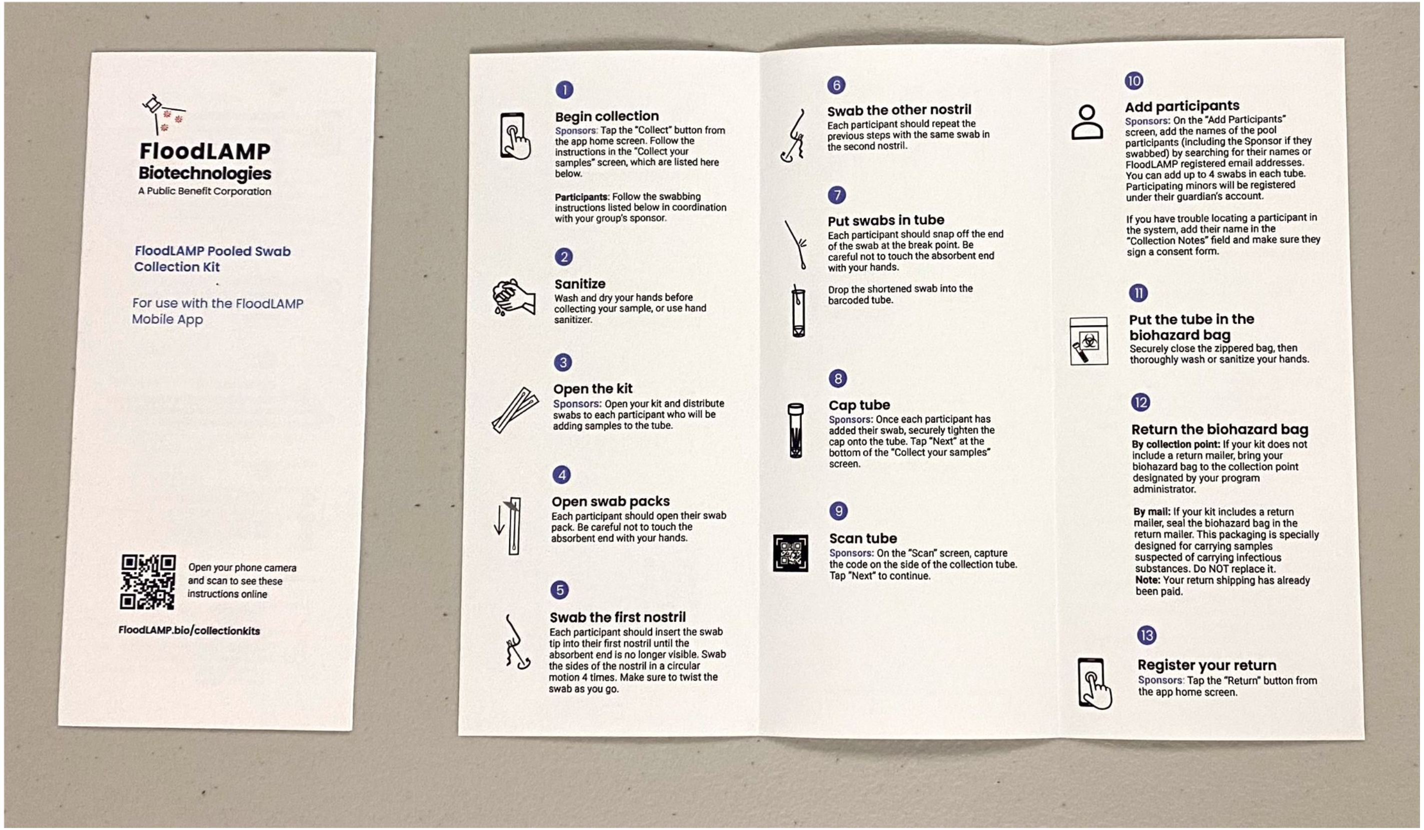
- All-in Test Kits
- Sample Processing Sites (Labs)
- Service, Support, Consulting
- Testing Program Management

# Reagent Kits


# Collection Kits

# Collection Instructions



# Equipment and Supplies

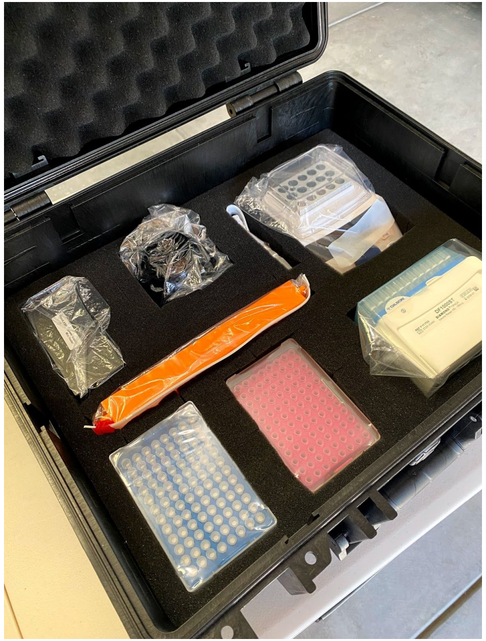

# "Lab"


DAVIE FIRE RESCUE
"PIT CREW"
Schematic for Cardiac Arrests
DO NOT THROW AWAY!!

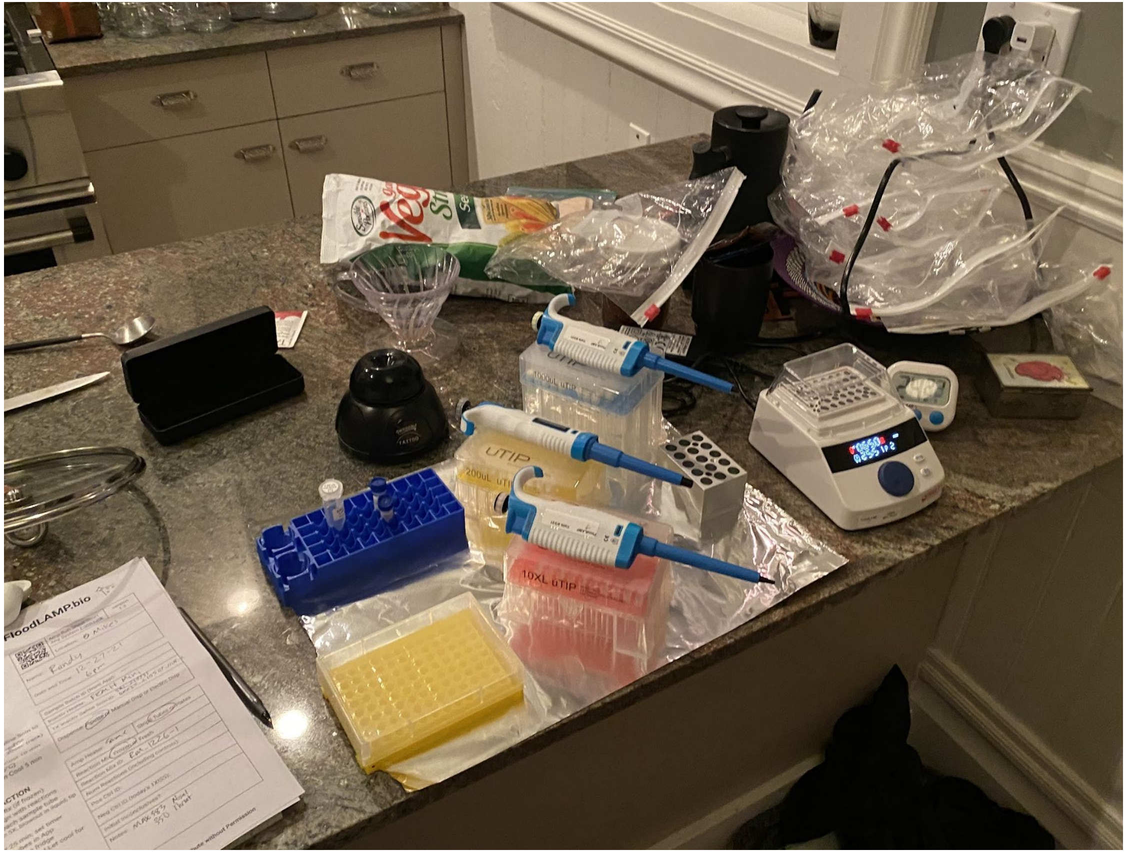

CLOROX
Purell
Ziploc

# Portable System



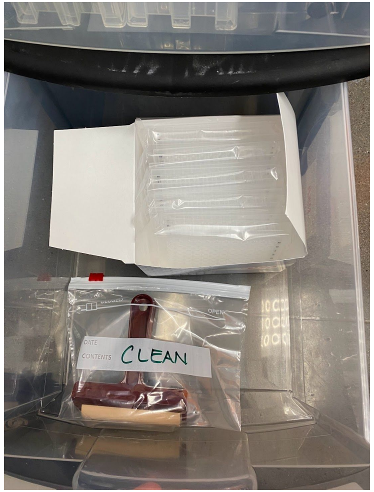


# Portable System - in action



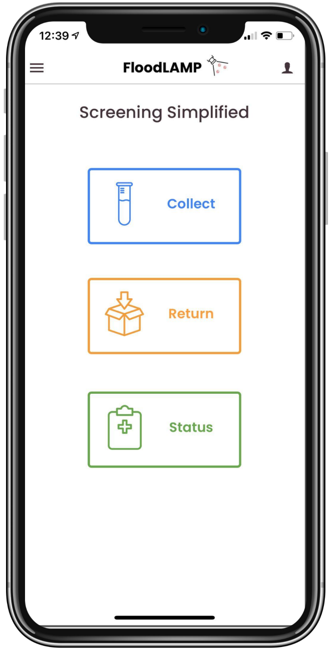
# Prepared Cart

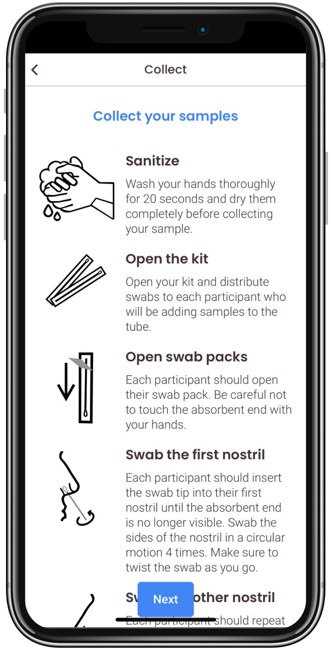


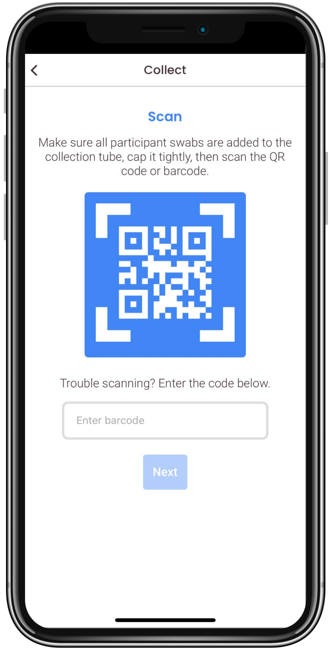


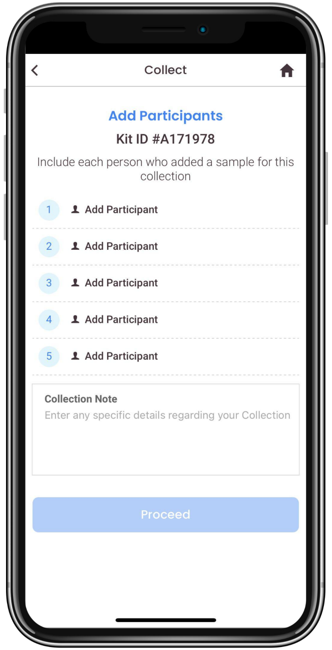
# App

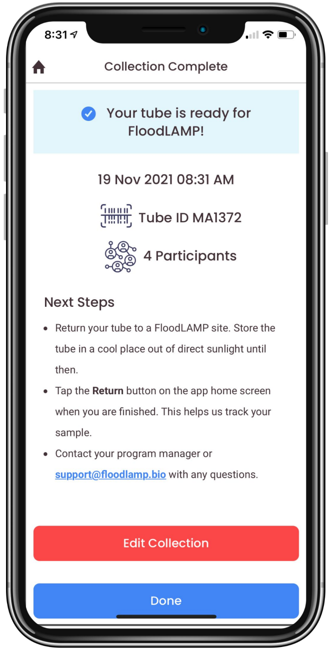
## For program participants



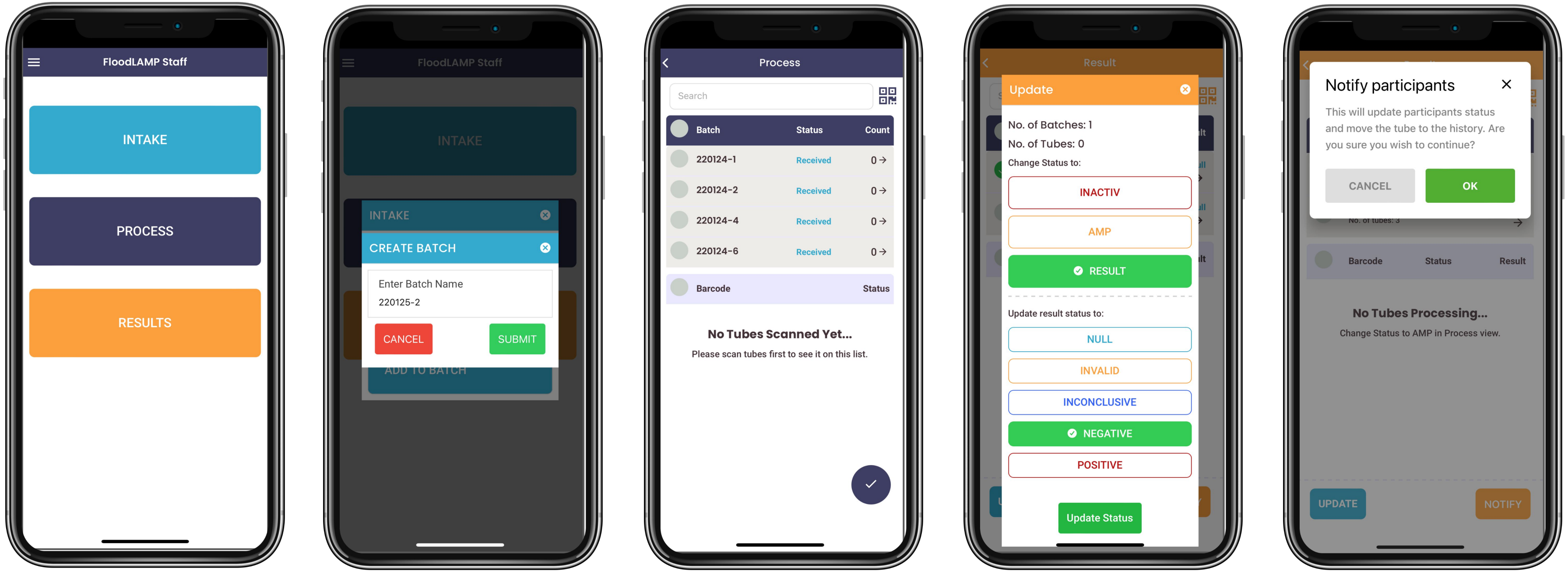



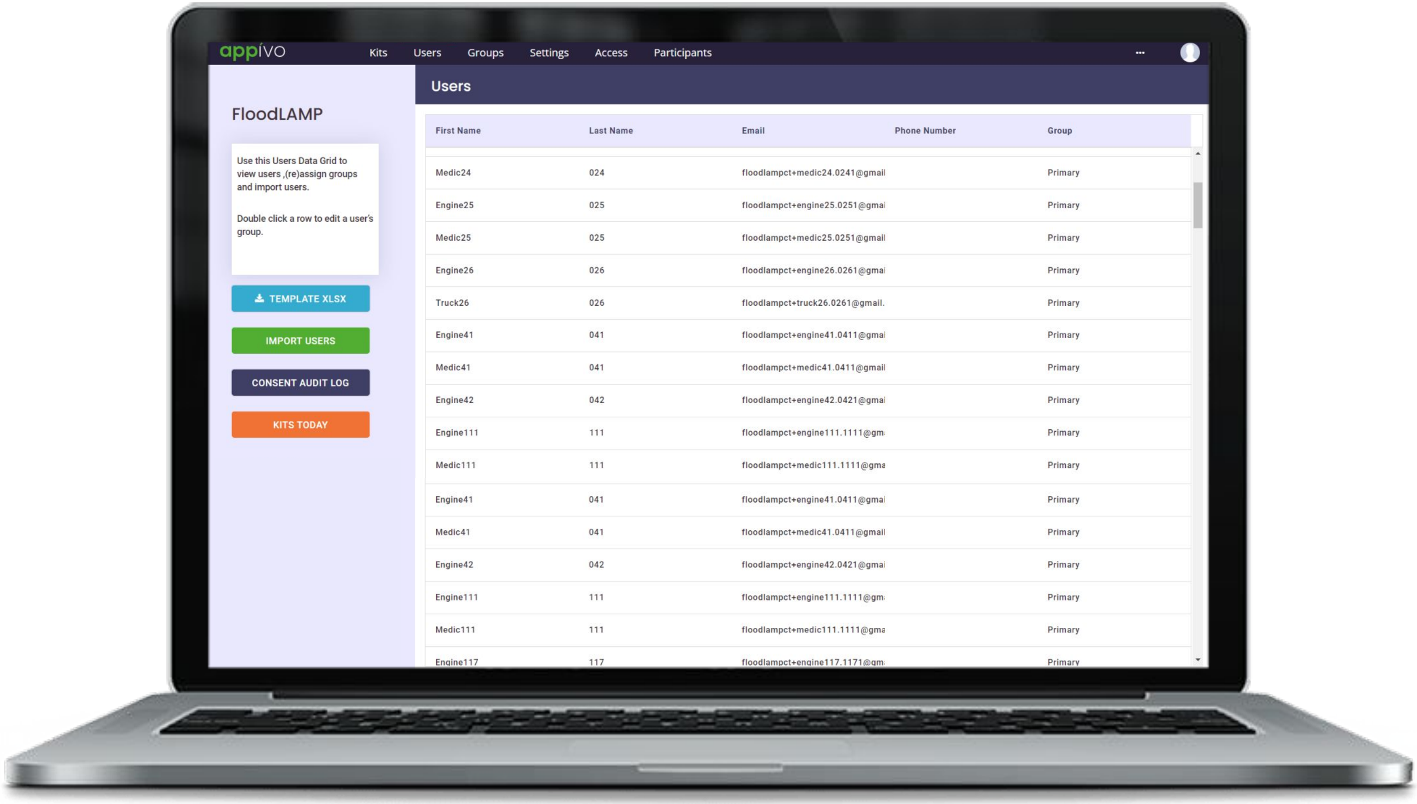
# App

## For the lab

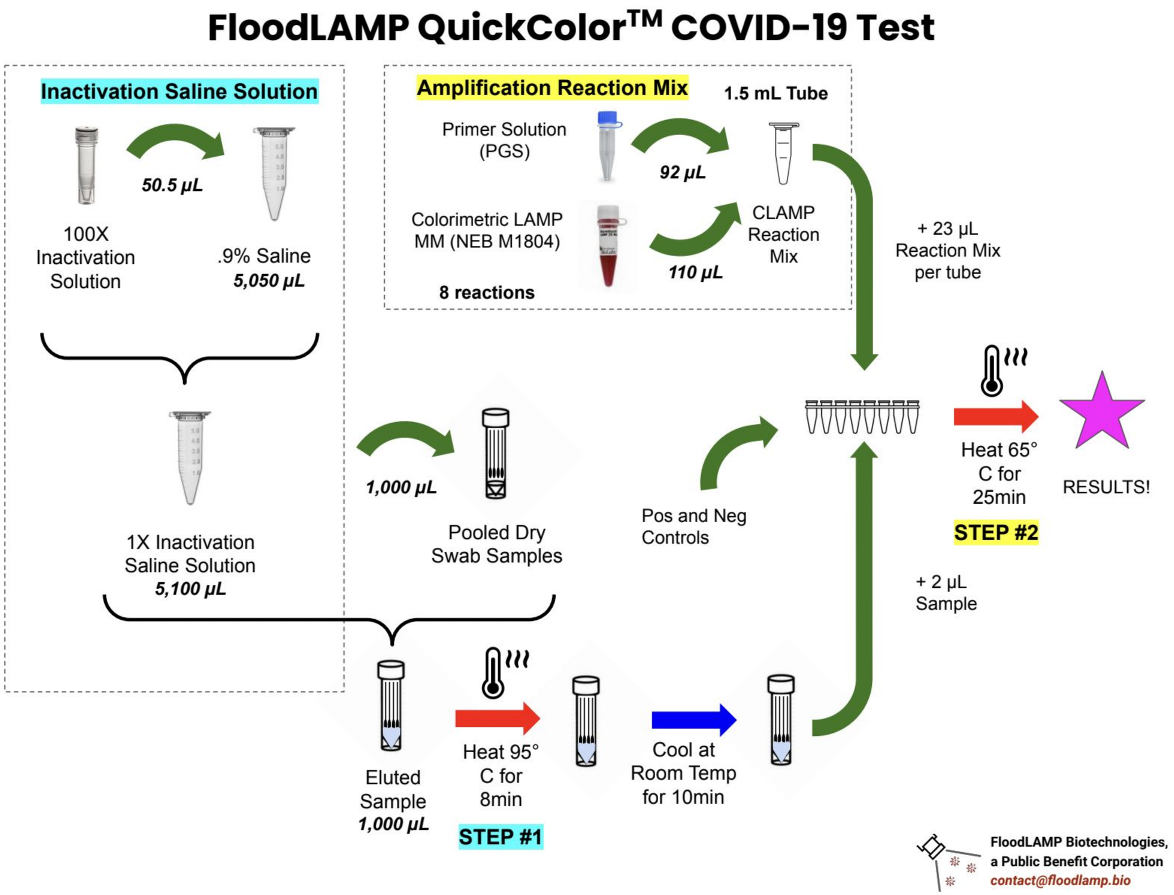


# Admin portal

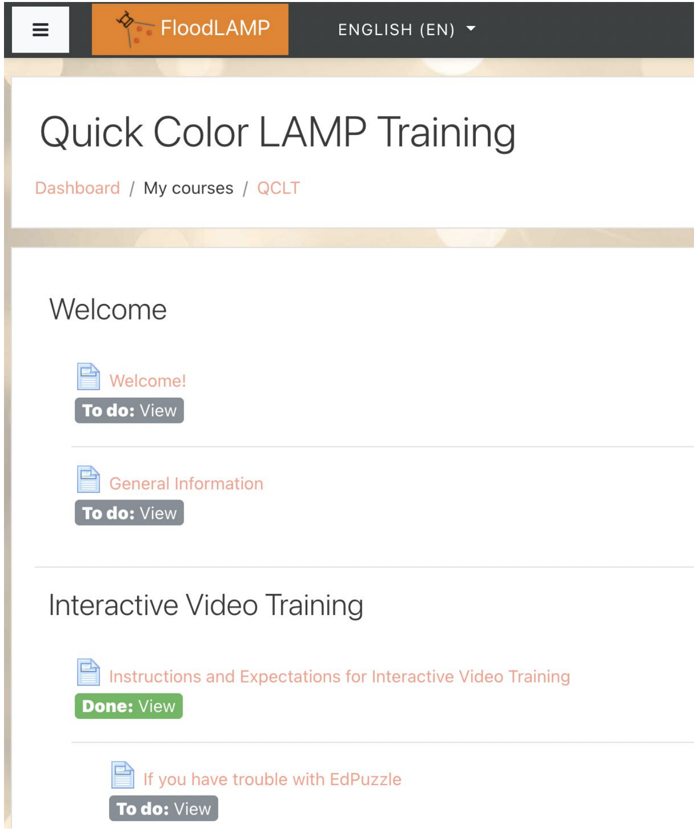
Tools for program administrators

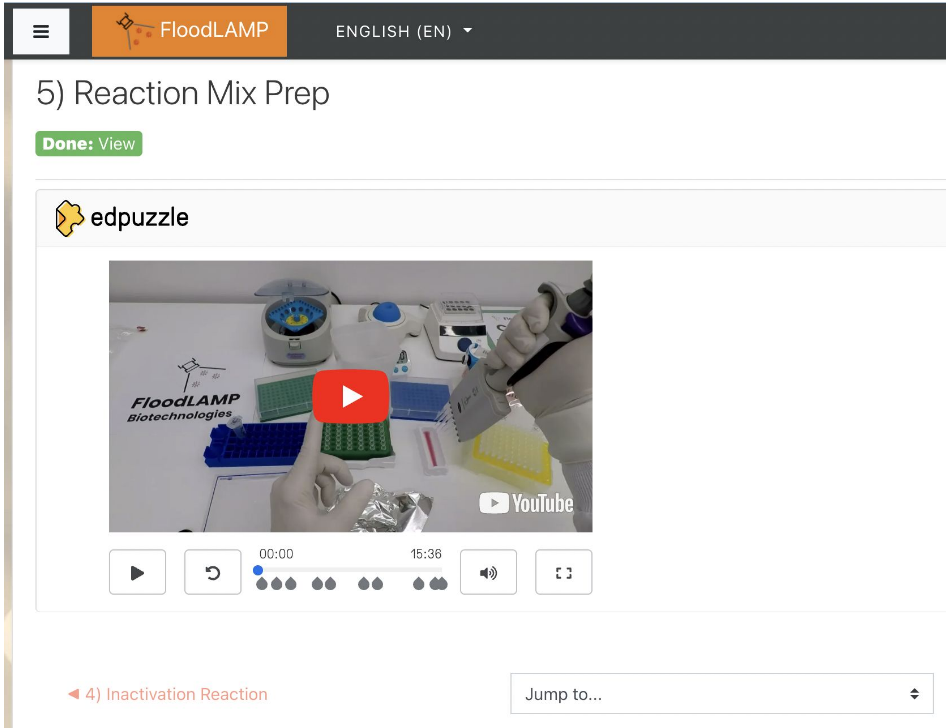


# Test Diagrams

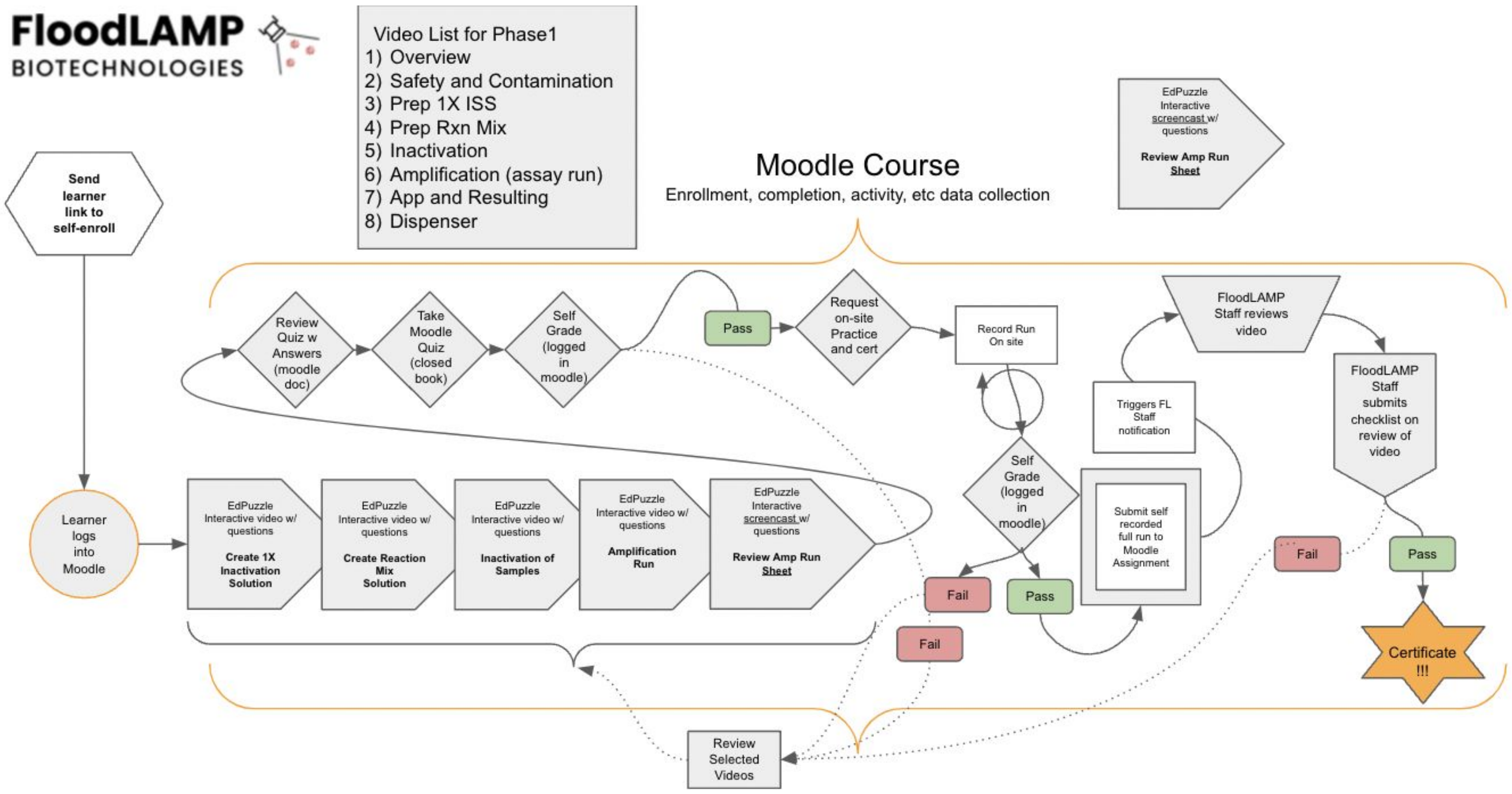


# Training Program – Interactive Video



# Training Program - Assessment for Certification



# Quality Management System

- Implementing ISO 13485 QMS
- QR Form Driven Logs

| FloodLAMP BIOTECHNOLOGIES | Document Number: | SOP-102-A_1 |
|---|---|---|
| | Effective Date: | 01/22/2022 |
| | Version: | 1 |
| | **Reaction Mix Prep - Table Form** | |

| | |
|---|---|
| CLAMP MM - NEB M1804<br>LOT:<br>EXP: | |
| New In Use CLAMP MM<br>Name:<br>Location:<br>Date and Time: | Retire Link<br>Name:<br>Date and Time: |

**Reaction Mix Prep**

1) Thaw PGS and CLAMP MM
2) Vortex PGS and CLAMP MM for 3s, centrifuge/shake down
3) Add CLAMP MM to PGS tube (or new 1.5mL tube for custom amount)
4) Vortex for 10s and centrifuge/shake down
5) Add 23µL of reaction mix per well {at bottom, no blowout, no tip touch}
   * optional use reservoir and multichannel
6) Check volumes (look at pink liquid) by lifting plate or strip tubes out of rack
7) Label with Reaction Mix ID

| | | | | | 48<br>8 | 550<br>92 | 655<br>110 |
|---|---|---|---|---|---|---|---|
| **Name** | **Date + Time** | **Reaction Mix ID (RM_MMDD-1)** | **CLAMP MM TubeID (_MMDD-X)** | **PGS LOT# + EXP** | **Num Rxns** | **PGS µL** | **CLAMP MM µL** |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

Version 1 SOP-102-A **Reaction Mix Prep - Table Form** Effective 22/01/2022

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| FloodLAMP BIOTECHNOLOGIES | Document Number: | SOP-103-A_1 |
|---|---|---|
| | Effective Date: | 01/22/2022 |
| | Version: | 1 |
| | **Amplification - Run Form** | |

**PREP**

- ☐ Heaters on (Heat indicator lit!)
- ☐ 1X Inactivation Saline Soln ready
- ☐ Reaction mix strip8/plates ready
- ☐ Safety Procedures:
  - lab coat
  - gloves
  - face mask
  - face shield or goggles
- ☐ Alcohol wipe sample tubes

| | |
|---|---|
| | Amplification Log Form Link |
| | Location: |
| Name: | |
| Date:<br>Time: | |

**INACTIVATION**

- ☐ Add 1mL of 1X Inactiv Saline Soln to each sample tube
- ☐ Vortex 10sec (tubes) or 30sec (rack)
- ☐ Water Bath (set 99°C)
  - Heat 8 min then Cool 10 min

OR

- ☐ Dry Heat Block (set 95°C)
  - Heat 5 min then Cool 5 min

| |
|---|
| Sample Batch ID (from App): |
| Inactiv Heater: |
| 1X Inactiv Saline Soln ID: |
| Dispense: Pipette or Manual Disp or Electric Disp |

**AMPLIFICATION REACTION**

- ☐ Thaw Reaction Mix (if frozen)
- ☐ Setup tips to align with reactions
- ☐ Add 2µL from each sample tube (pipet up& down 5X, blowout in liquid, tip touch)
- ☐ Heat 65°C for 25 min, set timer
- ☐ Intake sample tubes in App
- ☐ Put sample tubes in fridge
- ☐ Remove Reactions and Let cool for 1 min before photo
- ☐ Take photo in lightbox
- ☐ Crop photo and apply vivid filter
- ☐ Update results in App
- ☐ Log Run with Form link

| | |
|---|---|
| Amp Heater: | Strip8 Tubes or Plates |
| Reaction Mix: Frozen or Fresh | |
| Reaction Mix ID: | |
| Num Reactions (including controls): | |
| Pos Ctrl ID: | |
| Neg Ctrl ID (today's 1XISS): | |
| Initial Inconclusives?: | |
| Notes: | |

Version 1 SOP-103-A **Amplification - Run Form** Effective 22/01/2022

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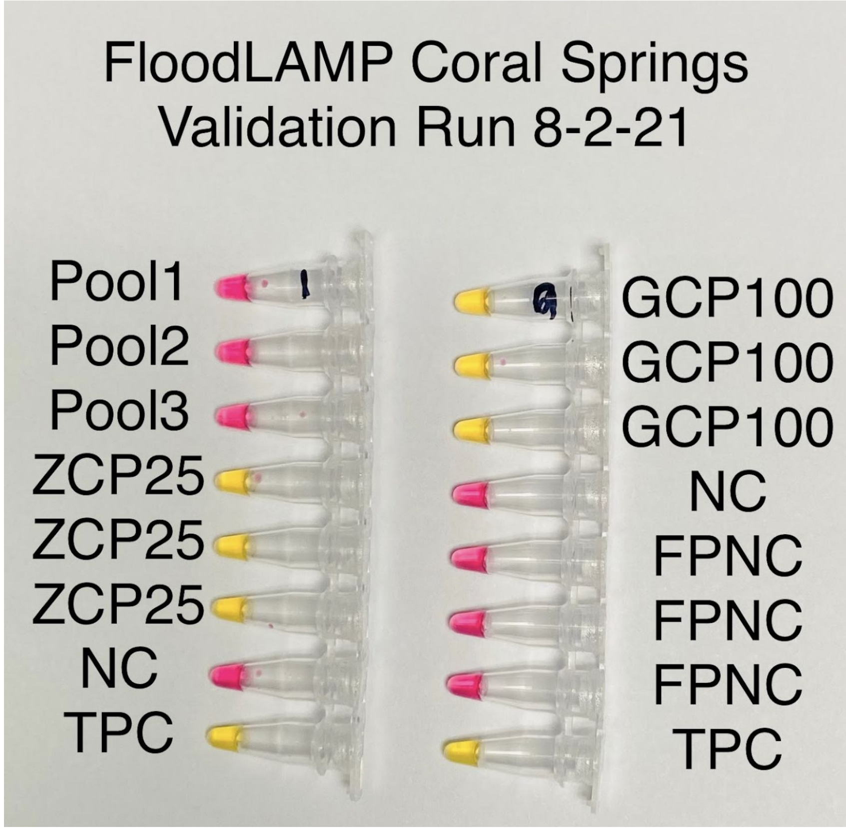
# New Site Validation Run



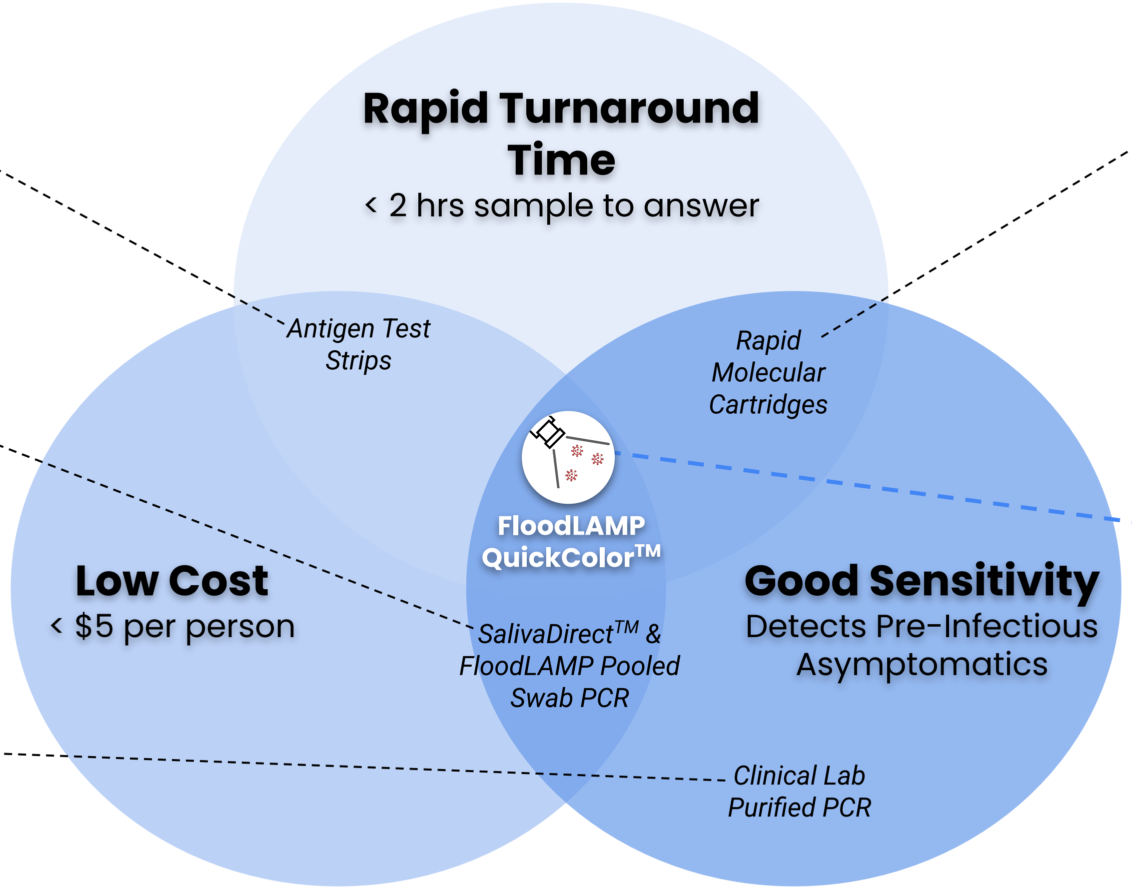
# 3) Open Source “Generic” Tests

# At Sweet Spot of Performance

People want higher confidence for both positive and negative results.

High upfront capital cost for PCR instruments limit replicability. These 2 tests have potential to drive down PCR costs.

Long turnaround renders it unsuitable for large scale screening - schools and workplaces need to know quickly if someone is infected.




Will help usher in era of at-home concierge diagnostics but simply too constrained by manufacturing and too expensive for public health screening.

- Improved performance
- Good sensitivity
- Fast turnaround
- Cheap
- Molecular
- Flexible
- Decentralised
- Resilient

# Filling a Gap in the Testing Landscape

At Home OTC | Point of Care | **Point of Need** | CLIA Labs

**Current testing paradigms cannot stem a pandemic caused by an asymptomatically transmitted, aerosolized pathogen.**

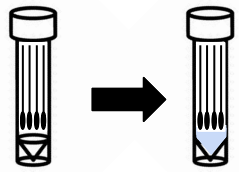
- Scaling a single test up to very high levels offers **opportunities for innovation** in sampling, test chemistry, and program configuration.
- A "**Point of Need**" modality of testing is needed that combines the scalability and low cost of liquid reagent processing with the ease, flexibility and decentralized nature of POC/At-Home.
- **Public Health processing sites** can fill the gap between CLIA labs and DIY at home tests.
- Want this new capability to be **additive** and not cannibalize the clinical diagnostic infrastructure.

# LAMP EUAs for SARS-CoV2

| EUA Date | Test | Diagnostic | Target | IC | Detection | Extraction | Amplification | LOD spike | LOD GE/ 3ml swab |
|---|---|---|---|---|---|---|---|---|---|
| Nov-17 | Lucira (At Home POC) | Lucira COVID-19 All-In-One Test Kit | N, N | External + IC | Colorimetric | ~Direct | Lucira device | Inactivated virus | 2,700 |
| Oct-05 | Seasun (2 duplex) | AQ-TOP COVID-19 Rapid Detection Kit PLUS | orf1ab, N | Human RNaseP | PNA Probe | Manual (Qiagen_60704 or Seasun_SS-1300) or Automated (Panagene_PNAK-1001 on PanaMax48) | BioRad_CFX96 or ABI7500 | NCCP_43326 genomic RNA | 3000 |
| Sep-01 | Detectachem | MobileDetect Bio BCC19 (MD-Bio BCC19) Test Kit | N, E | only external controls | Colorimetric | Direct (1µl of transport media into LAMP) | heat block or qPCR (MD-Bio heater or BioRad_T100 or AB_Veriti or ...) | Twist_MT007544.1 synthetic gRNA | 225,000 |
| Aug-31 | Mammoth | SARS-CoV-2 DETECTR Reagent Kit | N | Human RNaseP | CRISPR Probe | Automated (Qiagen_955134 on Qiagen_EZ1AdvancedBenchtop) | ABI7500 | SeraCare_AccuPlex_0505-0168 IVT encapsulated RNA | 60,000 |
| Aug-13 | Pro-Lab | Pro-AmpRT SARS-CoV-2 Test | RdRP | only external controls | Probe | Direct (swab into 0.1ml) or kit (swab into 1ml then Pro-Lab_PLM-2000) | Optigene_GenieHT | BEI_NR-52287 inactivated virus | 125* |
| Jul-09 | UCSF/Mammoth | SARS-CoV-2 RNA DETECTR Assay | N | Human RNaseP | CRISPR Probe | automated (Qiagen_955134 on Qiagen_EZ1) | ABI7500 | SeraCare_AccuPlex_0505-0126 (lot 10480311) IVT RNA | 60,000 |
| May-21 | Seasun (1 duplex) | AQ-TOP COVID-19 Rapid Detection Kit | orf1ab | Human RNaseP | PNA Probe | manual (Qiagen_60704) | BioRad_CFX96 or ABI7500 | NCCP_43326 gRNA | 21,000 |
| May-18/ Aug31 | Color | Color Genomics SARS-CoV-2 RT-LAMP Diagnostic Assay | N, E, nsp3 | Human RNaseP | Colorimetric | automated (PerkinElmer_CMG-1033 on PerkinElmer_Chemagic360) | plate reader (Biotek_NEO2), Hamilton_Star | ATCC_VR-1986D (gRNA) | 2,250 |
| May-06 | Sherlock BioSci | Sherlock CRISPR SARS-CoV-2 Kit | orf1ab, N | Human RNaseP | CRISPR Probe | Manual (ThermoFIsher_12280050) | heat block or qPCR, Plate reader (BioTek_NEO2) | gRNA | 20,250 |
| Apr-10 | Atila BioSystems | iAMP COVID-19 Detection Kit | orf1ab, N | human GAPDH | Probe | Direct (swab into 350µl, 15min RT then 3µl to LAMP) | BioRad_CFX96 or ABI7500 or Roche_LightCycler480II or Atila_PG9600 | SeraCare_AccuPlex_0505-0129 pseudovirus (recombinant alphavirus) | 3,500* |

**Just in 2020!**
**from Matt McFarlane**
**(twitter @mattmcfar)**

# PCR and LAMP Sister Assays



*visual readout*

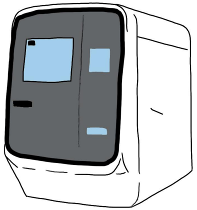
## Streamlined Sample Prep

- Standard Rabe-Cepko TCEP/EDTA inactivation solution
- Added directly to dry swabs
- Same sample for both tests

- Same sample for both tests
- 2μL sample volume

## QuickColor™ LAMP Test

- M1804 MM
- Standard by-the-book 25μL reaction
- 3 primer sets mixed (AS1e, N2, E1)
- 25 minutes at 65°C



*Standard RT-qPCR instrument*

## EasyPCR™ Test

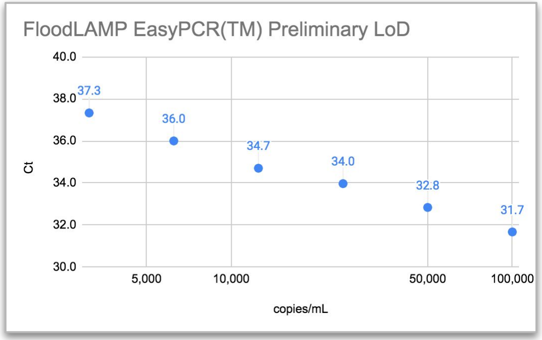
- Luna E3006 kit
- Standard by-the-book 20μL reaction
- CDC primers in SalivaDirect config (N1, RNAseP)

# Clinical Evaluation Data

Clinical evaluation performed by the Stanford CLIA Lab, with excellent results and praise on the "**really straightforward**" protocol.

## EasyPCR™ Test

- 3 copies/µl LoD
- 98% sensitivity (PPA 39/40)
- 100% Specificity (40/40)
- No false positives




Gamma inactivated cell lysate from BEI spiked into raw clinical negative sample

| FloodLAMP SwabDirect PCR Result | Comparator Positive | Comparator Negative | Total |
|---|---|---|---|
| Positive | 39 | 0 | 39 |
| Negative | 1 | 40 | 41 |
| Invalid | 0 | 0 | 0 |
| Total | 40 | 40 | 80 |
| Positive Agreement | 97.5% (39/40) 95% CI: 86.8% to 99.9% | | |
| Negative Agreement | 100% (40/40) 95% CI: 91.2% to 100% | | |

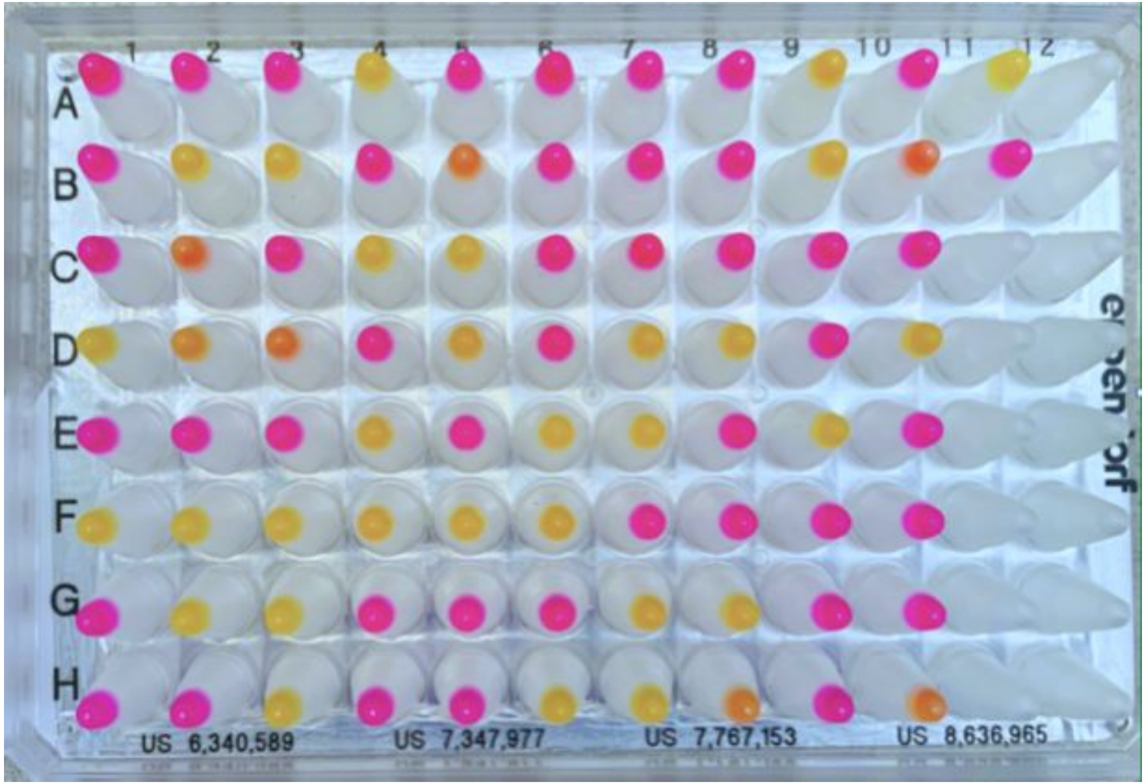
## QuickColor™ LAMP Test

- 12 copies/µl LoD
- 90% Sensitivity (PPA 36/40)
- Missed positives only high Ct (>36 with direct PCR)
- 100% Specificity (40/40)
- No false positives

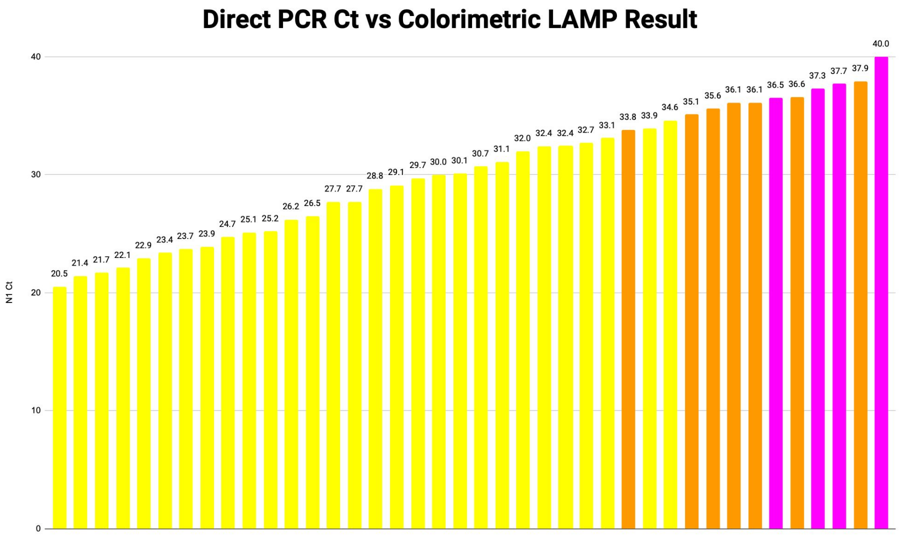
| FloodLAMP QuickColor Test Result | Comparator Positive | Comparator Negative | Total |
|---|---|---|---|
| Positive | 36 | 0 | 36 |
| Negative | 4 | 40 | 44 |
| Total | 40 | 40 | 80 |
| Positive Agreement | 90.0% (36/40) 95% CI: 76.3% to 97.2% | | |
| Negative Agreement | 100% (40/40) 95% CI: 91.2% to 100% | | |

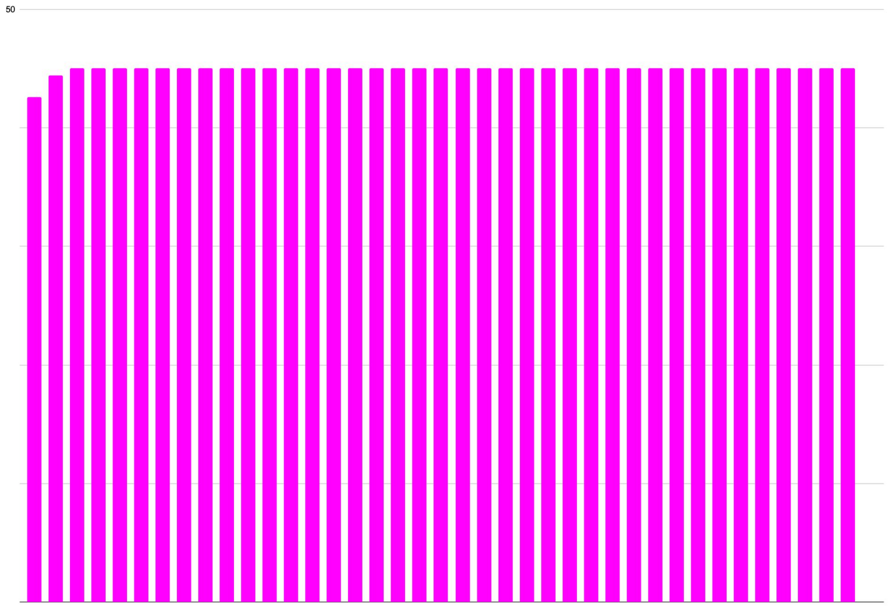
Source of Specimens: Stanford COVID-19 Clinical Testing Program
Specimen Type: Anterior Nares Swab in PBS, previously tested and frozen
Comparator Test: Hologic Panther Fusion SARS-CoV-2 Assay and Hologic Panther Aptima SARS-CoV-2 Assay

# Clinical Evaluation by Stanford CLIA Lab



# Clinical Evaluation by Stanford CLIA Lab

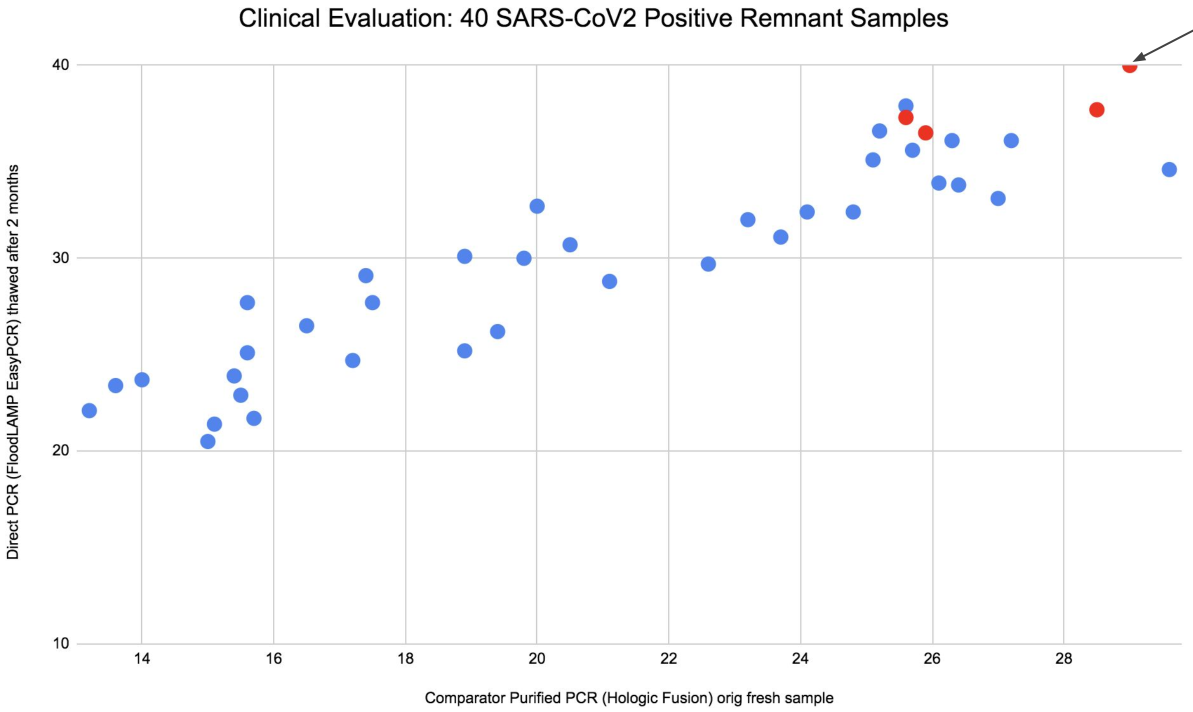



40 Clinical Positive Samples:
29 Positive by FloodLAMP QuickColor™ Test
7 Inconclusive (all high Ct, low viral load)
4 Negative (all > 36 Ct, very low viral load)

40 Clinical Negative Samples:
40 Negative by FloodLAMP QuickColor™ Test

# Clinical Evaluation by Stanford CLIA Lab



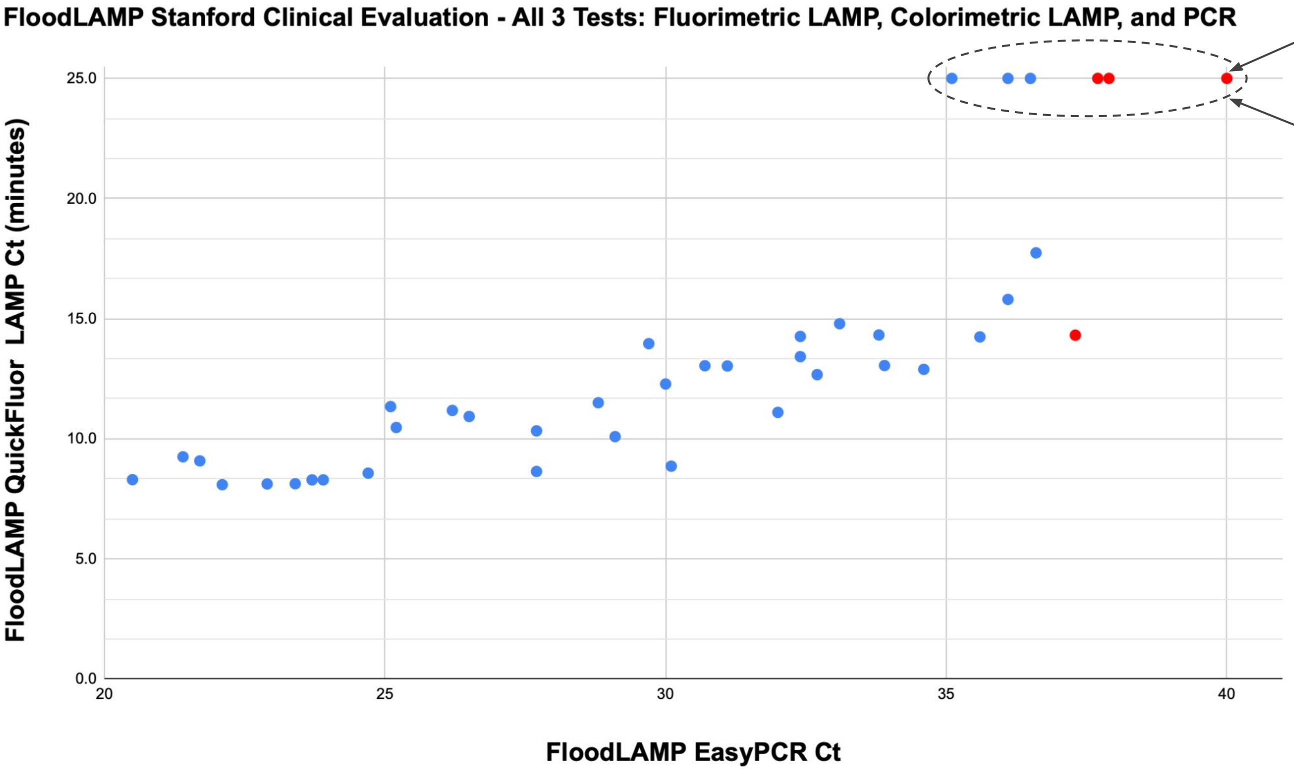

One sample Undetermined (not detected) by FloodLAMP™ EasyPCR

- Detected in FloodLAMP EasyPCR™ Not detected by FloodLAMP QuickColor™ (EasyPCR™ PPA 39/40 97.5%)

- Detected in both FloodLAMP EasyPCR™ and QuickColor™ Colorimetric LAMP test (QuickColor™ PPA 36/40 90%)
  * includes initial inconclusive resolved by PCR

Both FloodLAMP EasyPCR™ and QuickColor™ had no false positives (NPA 40/40 100%)

# Clinical Evaluation by Stanford CLIA Lab




One sample not detected by FloodLAMP™ EasyPCR (PPA 39/40 97.5%)

6 Positives Not Detected by FloodLAMP QuickFluor™ test (PPA 34/40 85%)

● Positives not detected by FloodLAMP QuickColor™ test (PPA 36/40 90%)
* includes initial inconclusives resolved by PCR

All 3 Tests:
FloodLAMP EasyPCR™
QuickColor™ LAMP
QuickFluor™ LAMP
had no false positives
(NPA 40/40 100%)

# Saliva Direct "most unique EUA" - FDA

- Is a protocol not a product
- Open Access - any CLIA can use
- Lots of clone assays being used

- Yale can "designate" CLIA labs

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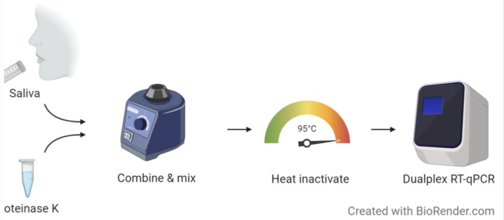
## SalivaDirect: What You Need to Know About the New COVID-19 Test

September 03, 2020 by Michael Greenwood




COVIDTracker

About our Test Development

**SalivaDirect™ has moved!**

For information regarding SalivaDirect™, including lab designation and up-to-date validation results, please visit our new page at https://publichealth.yale.edu/salivadirect/.

COVIDTrackerCT will remain home to our surveillance, open-source protocols, and research (including test development).

## SalivaDirect: Lab authorization request form

SalivaDirect received Emergency Use Authorization (EUA) from the Food and Drug Administration (FDA) on August 15th, 2020.

The SalivaDirect FDA EUA is for a Laboratory Developed Test, but we have the right to designate other labs its use.

Only high complexity CLIA-certified labs in the United States can become authorized to run SalivaDirect. Please fill out this form if you would like to receive more information on how to become a designated lab.

Do you have access to at least 5x SARS-CoV-2 positive and 5x negative saliva samples for CLIA verification? *

Would you be open to collaboration on bridging studies for your automation systems?

# Open EUA's - 2 Full Submissions + Pre EUA

## FloodLAMP QuickColor™ COVID-19 Test

Instructions for Use v1.2

IVD

COVID-19 Emergency Use Authorization Only
For *in vitro* diagnostic (IVD) Use

www.floodlamp.bio
FloodLAMP Biotechnologies, PBC | 930 Brittan Ave. San Carlos, CA 94070 USA

## FloodLAMP EasyPCR™ COVID-19 Test

Instructions for Use v1.1

IVD

COVID-19 Emergency Use Authorization Only
For *in vitro* diagnostic (IVD) Use

www.floodlamp.bio
FloodLAMP Biotechnologies, PBC | 930 Brittan Ave. San Carlos, CA 94070 USA

**FloodLAMP**
**Biotechnologies**
A Public Benefit Corporation

**FloodLAMP Pooled Swab Collection Kit DTC**

For use with the FloodLAMP Mobile App

# Open EUA - Disclosure of Components

FloodLAMP QuickColor™ COVID-19 Test
IVD FDA Emergency Use Authorization - Instructions for Use (v1.2)

**Table 1: Validated reagents used with the Test**

| Item | Concentration | Chemical Composition | Vendor | Catalog Number |
|---|---|---|---|---|
| TCEP | .5 M | tris(2-carboxyethyl)phosphine hydrochloride | Sigma-Aldrich Millipore Sigma | 646547-10X1ML |
| EDTA | .5 M | Ethylenediaminetetraacetic acid | Thermo Fisher | 15575020 |
| NaOH | 10 N | Sodium Hydroxide | Sigma-Aldrich | SX0607N-6 |
| Nuclease-free Water | | Ultrapure Water, nuclease-free | Thermo Fisher | 10977015 |
| NaCl | 5 M | Sodium Chloride | Thermo Fisher | 24740011 |
| Guanidine HCl | 6 M | Guanidine Hydrochloride | Sigma-Aldrich | SRE0066 |
| Colorimetric LAMP MM* | | Colorimetric LAMP Master Mix | New England Biolabs | M1804 |

* Item may not be substituted for equivalent. Only the specified vendor and catalog number may be utilized.

The FloodLAMP QuickColor™ COVID-19 Test uses 18 LAMP primers targeted for 3 different SARS-CoV-2 genes, with 6 primers for each target. All 18 primers are mixed together and are input into a single amplification reaction. Primer names and sequences are listed in Table 2. Primers may be purchased pre-blended from the vendor LGC Biosearch Technologies with the product names LAMP_S2-As1e, LAMP_S2-N2, LAMP_S2-E1. Alternatively, primers may be purchased as individual custom oligos. Appropriate validation of primer mixes from custom oligos is required. See Primer Preparation below for more information.

**Table 2: Primer names and sequences**

| Primer Name | Sequence (5'-3') |
|---|---|
| **ORF1ab gene (AS1e)** | |
| Orf1ab_FIP | TCAGCACACAAAGCCAAAAATTTATTTTTCTGTGCAAAGGAAATTAAGGAG |
| Orf1ab_BIP | TATTGGTGGAGCTAAACTTAAAGCCTTTTCTGTACAATCCCTTTGAGTG |
| Orf1ab_F3 | CGGTGGACAAATTGTCAC |
| Orf1ab_B3 | CTTCTCTGGATTTAACACACTT |
| Orf1ab_LF | TTACAAGCTTAAAGAATGTCTGAACACT |
| Orf1ab_LB | TTGAATTTAGGTGAAACATTTGTCACG |
| **N Gene (N2)** | |
| N2_FIP | TTCCGAAGAACGCTGAAGCGGAACTGATTACAAACATTGGCC |
| N2_BIP | CGCATTGGCATGGAAGTCACAATTTGATGGCACCTGTGTA |
| N2_F3 | ACCAGGAACTAATCAGACAAG |
| N2_B3 | GACTTGATCTTTGAAATTTGGATCT |
| N2_LF | GGGGGCAAATTGTGCAATTTG |
| N2_LB | CTTCGGGAACGTGGTTGACC |

**Table 7: Primer-Guanidine Solution**

| Component | Volume (1 reaction) | Volume (1 reaction x 100) 1 x 96-plate w/ 4% overage |
|---|---|---|
| 10X LAMP Primer Mix | 2.5 µL | 250 µL |
| Guanidine HCl (400 mM) | 2.5 µL | |
| Guanidine HCl (6 M) | | 16.7 µL |
| Nuclease-free Water | 5.5 µL | 783 µL |
| **TOTAL VOLUME** | **10.5 µL** | **1050 µL** |

**Table 8: Colorimetric LAMP Amplification Reaction**

| Component | Volume (1 reaction) | Volume (100 reactions) |
|---|---|---|
| Primer-Guanidine Solution | 10.5 µL | 1050 µL |
| Colorimetric LAMP MM | 12.5 µL | 1250 µL |
| **SUBTOTAL VOLUME** | **23 µL** | **2300 µL** |
| Sample | 2 µL | |
| **REACTION VOLUME** | **25 µL** | |

# Open EUA's

Provides a path to **establishing generics in diagnostics**.

| | Typical IVD EUA | CDC EUA | SalivaDirect™ | SHIELD | FloodLAMP EasyPCR™ | FloodLAMP QuickColor™ |
|---|---|---|---|---|---|---|
| Disclosure of all chemicals and reagents? | No | Yes | Yes | Yes | Yes | Yes |
| Chemical and reagents available from multiple vendors? | No | Yes | Yes | No | Yes | Yes |
| Disclosure of primer sequences? | No | Yes<br>std for PCR | Yes | No<br>ProptryThermo | Yes | Yes |
| Primers commercially available from multiple vendors? | No | Yes | Yes<br>CDC Primers | No<br>ProptryThermo | Yes<br>CDC SD Primers | Yes<br>Available but not launched |
| Supply chain robust? | No | No/Maybe | Yes | No/Maybe | Yes | Yes |
| EUA Sponsor Organization Type | For Profit Company | Govt | Academic Not for Profit | Academic Not/For Profit ? | Public Benefit Corp | Public Benefit Corp |
| Designation of CLIA labs | Kit Sales | N/A<br>open RoR | Impact & Expansion | Impact & Expansion | Impact & Expansion | Impact & Expansion |

# Push for Regulatory Progress

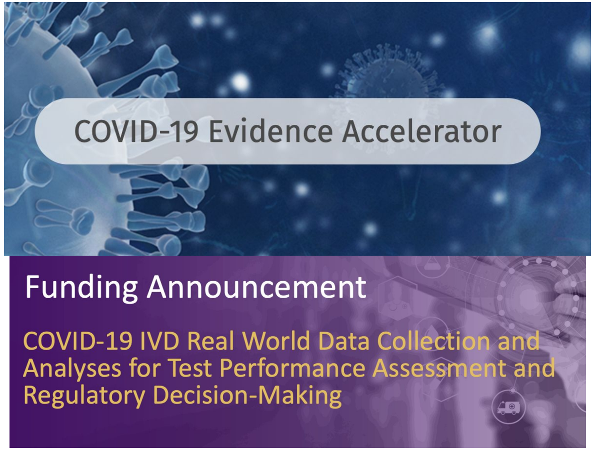
**Prepare for and Respond to Existing Viruses, Emerging New Threats, and Pandemics Act (PREVENT Pandemics Act)**

| | |
|---|---|
| Sec.505.Faciliating the use of real world evidence. | • Requires FDA to issue or revise guidance on the use of real-world data and real-world evidence to support regulatory decisionmaking, including with respect to real-world data and real-world evidence from products authorized for emergency use. |
| Sec.506.Advanced platform technologies. | • Creates an advanced platform technology designation to expedite the development and review of new treatments and countermeasures that use cutting-edge, adaptable platform technologies that can be incorporated or used in more than one drug or biological product.<br>• Requires FDA to issue guidance on the implementation of the new designation. |
| Sec.507.Increasing EUA decision transparency. | • Provides FDA with authority to share more safety and effectiveness information with the public about products authorized for emergency use. |
| Sec.508.Improving FDA guidance and communication. | • Requires publication of a report identifying best practices across the FDA and other applicable agencies for the development, issuance, and use of guidance documents and for communications with product sponsors and other stakeholders, and a plan for implementing such best practices.<br>• Requires FDA to publish a report on the agency's best practices for communicating with medical product sponsors and other stakeholders, and a plan for implementing such best practices. |
| Sec.509.GAO study and report on hiring challenges at FDA. | • Directs GAO to issue a report assessing FDA's hiring, recruiting, and retention practices, policies, and processes, and their impact on FDA's ability to carry out its public health mission, particularly in light of the COVID-19 pandemic. |

*Subtitle B—Mitigating Shortages*

https://www.help.senate.gov/imo/media/doc/PREVENT%20Pandemics%20discussion%20draft%20sxs%20final.pdf

REAGAN-UDALL FOUNDATION for the Food and Drug Administration




https://reaganudall.org/research-funding-opportunity

# Team Effort!



# FloodLAMP Team – Thank You!

**Randy True**
Founder & CEO

- *Founder TMI Inc. - Acq. Affymetrix*
- *Former VP of R&D at Affymetrix*

*bit.ly/biosketch-randalltrue*
*co-author global LAMP consortium review: bit.ly/gLAMP-review*
*https://abrf.memberclicks.net/jbt-2021-september-issue*

**Theresa Ling**
UX/Design Lead

**Dr. Peter Antevy, MD**
Program Medical Director

**Gary Withey, Ph.D.**
Vice President of Research & Development

**Brandon Smith**
Lab Assistant

## Scientific Advisory Board

**Anne Wyllie**
*Yale, Lead Researcher – SalivaDirect*
**Bill Hyun**
*UC San Francisco, Genoa Ventures*

## Industry Advisors

**Tim Lugo**
*William Blair Biotechnology Group Head*
**Zarak Khurshid**
*Asymmetry Capital, Top MDx Analyst*
**John Edge**
*Oxford Internet Institute, Blue Field Labs, ID2020*

## Acknowledgments

Antanas Sadunas, Sam Fogelberg, Carey Tan, Esmé Thornhill-Davis, Jeff Huber, Cliff Wang, Chris Mason, Simon Johnson, Brett Johnson, Mike Finney, Ron Cook
**NSVD:** Vincent Law, Michael Wells, Katrina Brandis, Gaby Hartley, Tim Earley, Nasreen Haque
**Research-Aid Networks:** Jeremy Rossman
**EMS deployments:** Adam, Danny, Alex, Petar, Dr. Paul Pepe
*Thanks too to all of our advisors, volunteers & former employees!!*

# APPENDIX

# Successful Commercial Pilots

- Commercial deployments in 3 states.
- On-site, rapid mass molecular testing performed and operated by EMS staff.
- 2 cities have opted for mandatory FloodLAMP testing. Prioritizing it over antigen and PCR alternatives.

"FloodLAMP's testing program is a total game-changer. I didn't know this was possible. I've been blown away."

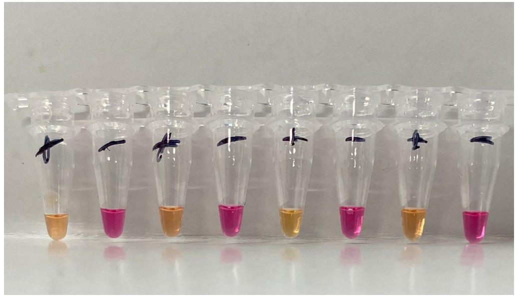
*– President of Production Company for FloodLAMP Pilot program "ROSA"*

"It's been a godsend!"

*– EMS Training Director running the FloodLAMP program*

# Collaboration with University of Antioquia (Medellin, Colombia)

Joint NEB / FloodLAMP project



The New York Times

HEALTH

# Several common rapid antigen tests work well for Omicron, according to a new study.

The new findings are from an ongoing U.S. study that began in October and was designed to assess the performance of rapid antigen tests in asymptomatic people.

By Emily Anthes

March 1, 2022


medRxiv

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CSH Cold Spring Harbor Laboratory BMJ Yale

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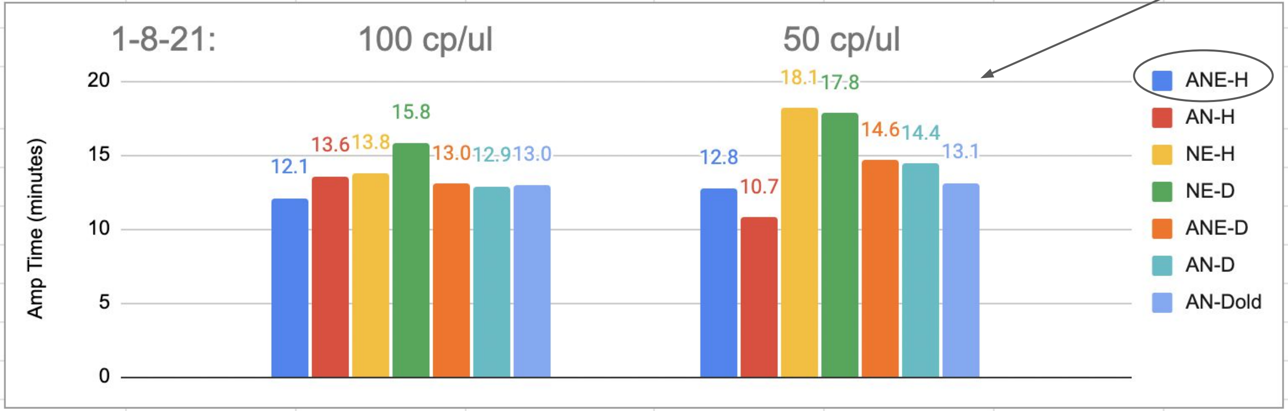
## Comparison of Rapid Antigen Tests' Performance between Delta (B.1.61.7;AY.X) and Omicron (B.1.1.529; BA1) Variants of SARS-CoV-2: Secondary Analysis from a Serial Home Self-Testing Study

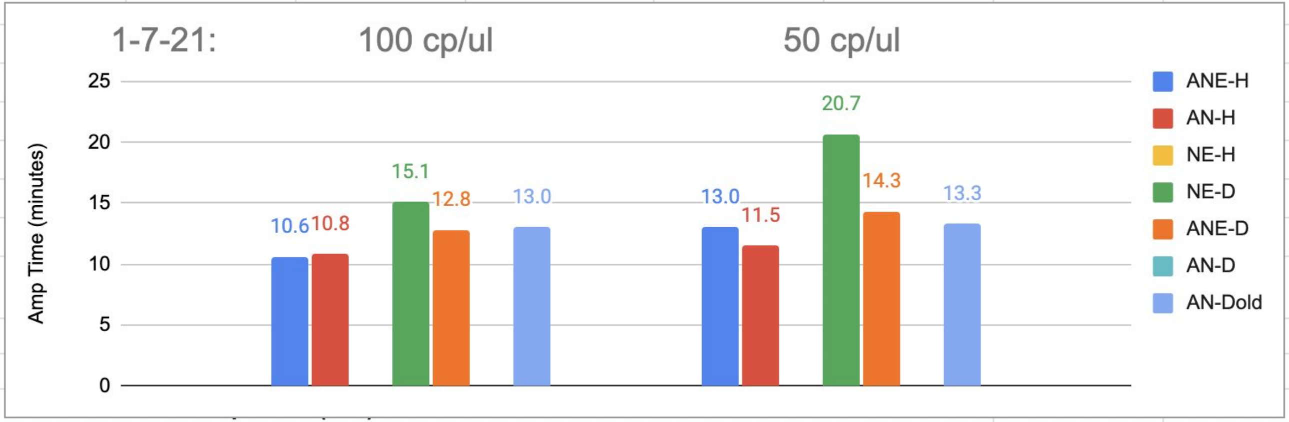
Apurv Soni, Carly Herbert, Andreas Filippaios, John Broach, Andres Colubri, Nisha Fahey, Kelsey Woods, Janvi Nanavati, Colton Wright, Taylor Orwig, Karen Gilliam, Vik Kheterpal, Thejas Suvarna, Chris Nowak, Summer Schrader, Honghuang Lin, Laurel O'Connor, Caitlin Pretz, Didem Ayturk, Elizabeth Orvek, Julie Flahive, Peter Lazar, Qiming Shi, Chad Achenbach, Robert Murphy, Matthew Robinson, Laura Gibson, Pamela Stamegna, Nathaniel Hafer, Katherine Luzuriaga, Bruce Barton, William Heetderks, Yukari C. Manabe, David McManus

**doi:** https://doi.org/10.1101/2022.02.27.22271090

**Results** From the 7,349 participants enrolled in the parent study, 5,506 met the eligibility criteria for this analysis. A total of 153 participants were RT-PCR+ (61 Delta, 92 Omicron); among this group, 36 (23.5%) tested Ag-RDT+ on the same day and 36 (23.5%) tested Ag-RDT+ within 48 hours as first RT-PCR+. The differences in sensitivity between variants were not statistically significant (same-day: Delta 16.4% [95% CI: 8.2-28.1] vs Omicron 28.2% [95% CI: 19.4-38.6]; and 48-hours: Delta 45.9% [33.1-59.2] vs.

# Primer Comparison




**All 3 primer sets chosen by HPLC purification**

Run with NEB E1700 Fluorimetric LAMP on QuantStudio7

*Sample is contrived positive with Gamma BEI spiked into negative clinical sample*

**A** = AS1e Rabe-Cepko primer set for ORF1ab

**N** = N2 primer set

**E** = E1 primer set

**"H"** = HPLC purified

**"D"** = Desalted